# GR-RHS: Grouped Regularized Horseshoe with a Group-Specific Allowance Layer

## 1 Methods

### 1.1 Grouped sparse regression framework

We consider a grouped sparse regression problem in which the coefficient vector is partitioned into known groups,

$$\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_G^\top)^\top, \qquad \boldsymbol{\beta}_g \in \mathbb{R}^{p_g}, \qquad \sum_{g=1}^{G} p_g = p.$$

Equivalently,

$$\{1, \dots, p\} = \mathcal{G}_1 \sqcup \cdots \sqcup \mathcal{G}_G, \qquad |\mathcal{G}_g| = p_g,$$

and

$$\boldsymbol{\beta}_g = (\beta_j : j \in \mathcal{G}_g).$$

Let $X_g \in \mathbb{R}^{n \times p_g}$ denote the design block associated with group $g$, and write $X = (X_1, \dots, X_G)$. The working regression model is

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_n), \tag{1}$$

where $\sigma^2 > 0$ denotes the noise variance.

The goal is to construct a grouped shrinkage prior with four features. First, the prior should retain the global-local backbone of the regularized horseshoe, so that small coefficients are strongly shrunk while large coefficients remain protected by a finite slab. Second, the hierarchy should distinguish between *group-level retention* and *within-group modulation*. Third, the construction should reduce to the standard regularized horseshoe in the one-coordinate limit. Fourth, the prior should expose a single group-level control parameter whose induced marginal behavior can be analyzed sharply.

The central new quantity is the *group-specific allowance parameter* $\kappa_g \in (0, 1)$, which governs how much signal group $g$ is allowed to retain after shrinkage.

### 1.2 Grouped regularized horseshoe prior

For each group $g$, introduce a latent allowance parameter

$$\kappa_g \in (0, 1),$$

and define the equivalent slab-scale parameter

$$c_g^2 := \frac{\sigma^2 \kappa_g}{1-\kappa_g}, \qquad \text{equivalently} \qquad \kappa_g = \frac{c_g^2}{\sigma^2 + c_g^2}. \tag{2}$$

Thus $\kappa_g$ and $c_g^2$ are in one-to-one correspondence. We work primarily with $\kappa_g$, since its unit-interval scale carries a direct allowance interpretation, while $c_g$ is often more convenient in analytical calculations.

For each coordinate $j \in \mathcal{G}_g$, define

$$\beta_j \mid \lambda_j, a_g, \kappa_g, \tau, \sigma^2 \sim \mathcal{N}(0, v_{j,g}), \tag{3}$$

where

$$v_{j,g} := \frac{\sigma^2 \kappa_g \, \tau^2 \lambda_j^2 a_g^2}{\sigma^2 \kappa_g + (1-\kappa_g)\tau^2 \lambda_j^2 a_g^2} = \frac{c_g^2 \tau^2 \lambda_j^2 a_g^2}{c_g^2 + \tau^2 \lambda_j^2 a_g^2}. \tag{4}$$

The hierarchy is completed by

$$\lambda_j \sim \mathrm{C}^+(0,1), \qquad a_g \sim \mathrm{HN}(0, s_{a,g}^2), \qquad \kappa_g \sim \Pi_{0,g}, \qquad \tau \mid \sigma \sim \mathrm{C}^+\big(0, \tilde{\tau}_0 \, \sigma\big), \qquad \sigma^2 \sim \Pi_\sigma, \tag{5}$$

where $\mathrm{C}^+(0,\cdot)$ denotes a half-Cauchy prior and $\mathrm{HN}(0,s^2)$ denotes the law of $|Z|$ for $Z \sim \mathcal{N}(0,s^2)$. Unless stated otherwise, the local scales $\lambda_j$, the group-amplitude parameters $a_g$, the allowance parameters $\kappa_g$, the global scale $\tau$, and the noise variance $\sigma^2$ are taken to be a priori independent up to the explicit conditioning on $\sigma$ in (5).

A fully Bayesian specification is obtained by supplementing the hierarchy with a prior $\Pi_\sigma$ on $\sigma^2$. In the present methods section and in the marginal prior theory developed below, however, $\sigma^2$ is often treated as fixed whenever this sharpens the structural interpretation.

This hierarchy preserves the global-local mechanism of the regularized horseshoe. The global parameter $\tau$ determines the baseline shrinkage level across all coordinates, the local parameters $\lambda_j$ allow coordinate-specific escape, and the finite slab enters through the group-level scale $c_g$. Relative to the standard regularized horseshoe, the new element is that the slab width is no longer common across coefficients; instead, it is group-specific and parameterized through the allowance variable $\kappa_g$.

It is important to distinguish conceptual role from posterior identifiability. The variance in (4) depends on the product $\tau a_g \lambda_j$, so the latent scale decomposition is not intended to be interpreted as a collection of fully orthogonal posterior objects. The layered hierarchy is instead introduced so that each component is assigned a primary structural role in the prior construction.

## 1.3 Interpretation of the hierarchy

Each layer of the hierarchy is assigned a primary role.

**Global layer.** The global scale $\tau$ controls the baseline sparsity level across all coordinates. Smaller values induce stronger global shrinkage, whereas larger values allow more coefficients to escape.

**Local layer.** The local half-Cauchy scales $\lambda_j$ retain the standard horseshoe escape mechanism: most coordinates are aggressively shrunk, but a subset may avoid strong regularization because of the heavy tails of the local prior.

**Group-amplitude layer.** The parameter $a_g$ acts as a common within-group amplitude modulator. Conditional on the group-level ceiling determined by $\kappa_g$, the pair $(a_g, \lambda_j)$ determines how closely each coordinate approaches the common variance ceiling shared by its group.

**Allowance layer.** The parameter $\kappa_g$ is the primary group-level gate. From (4),

$$v_{j,g} \leq c_g^2 = \frac{\sigma^2 \kappa_g}{1 - \kappa_g}, \qquad j \in \mathcal{G}_g. \tag{6}$$

Thus $c_g^2$ provides a common variance ceiling for the coordinates in group $g$, and $\kappa_g$ is the corresponding normalized group-level allowance ceiling. This is the reason for the term *group-specific allowance parameter*.

The central methodological viewpoint of the paper is that $\kappa_g$ should not be interpreted merely as an additional hyperparameter. It is the main group-level control object governing how much signal an entire group is allowed to retain after shrinkage.

## 1.4 Connection to the standard regularized horseshoe

The standard regularized horseshoe modifies the original horseshoe by replacing the infinitely wide slab with a finite slab width $c$, yielding an effective variance of the form

$$\tau^2 \tilde{\lambda}_j^2 = \tau^2 \frac{c^2 \lambda_j^2}{c^2 + \tau^2 \lambda_j^2}.$$

This preserves strong shrinkage near zero while soft-truncating the tails of the prior, thereby regularizing even large coefficients when the likelihood is weakly informative.

Our grouped prior retains exactly this logic. Rewriting (4),

$$v_{j,g} = \tau^2 \left( \frac{c_g^2 \lambda_j^2 a_g^2}{c_g^2 + \tau^2 \lambda_j^2 a_g^2} \right),$$

shows that the present method is a grouped regularized horseshoe in which the slab width becomes group-specific and the within-group local scale is $\lambda_j a_g$.

A useful reduction occurs when $p_g = 1$ and $a_g \equiv 1$. Then

$$v_{j,g} = \frac{c_g^2 \tau^2 \lambda_j^2}{c_g^2 + \tau^2 \lambda_j^2},$$

which is exactly the variance form of the standard regularized horseshoe with slab width $c_g$. Hence, under the specialization $p_g = 1$ and $a_g \equiv 1$, the primitive GR-RHS hierarchy reduces exactly to the one-coordinate regularized horseshoe with slab width $c_g$.

## 1.5 Grouped normal-means representation

The grouped regression model above defines the method in its intended general form. To make the shrinkage mechanism induced by the prior fully explicit, it is useful to study the grouped normal-means representation

$$Y_j \mid \beta_j \sim \mathcal{N}(\beta_j, \sigma^2), \qquad j = 1, \ldots, p, \tag{7}$$

where the coordinates are partitioned into groups

$$\{1, \ldots, p\} = \mathcal{G}_1 \sqcup \cdots \sqcup \mathcal{G}_G, \qquad |\mathcal{G}_g| = p_g.$$

This representation is introduced not because the method is restricted to normal means, but because it exposes the group-level allowance geometry in an analytically transparent form. It also makes the conditional shrinkage mechanism explicit and yields the key retention inequality used throughout the interpretive development below.

## 1.6 Marginal shrinkage interpretation under grouped normal means

The present hierarchy is most naturally interpreted through its induced shrinkage mechanism rather than through a closed-form marginal prior density for $\beta_j$. Its organizing principle is a *marginal shrinkage interpretation*: the group-specific allowance parameter $\kappa_g$ determines the maximal level of retention available to group $g$, while the local scales $\lambda_j$ and the group-amplitude parameter $a_g$ determine how closely each coordinate approaches that group-level ceiling.

This interpretation is already visible at the variance level. For $j \in \mathcal{G}_g$,

$$v_{j,g} = \frac{c_g^2 \tau^2 \lambda_j^2 a_g^2}{c_g^2 + \tau^2 \lambda_j^2 a_g^2} \leq c_g^2,$$

where $c_g^2 = \sigma^2 \kappa_g / (1 - \kappa_g)$. Thus, conditional on $\kappa_g$, every coordinate in group $g$ is regularized under the common variance ceiling $c_g^2$. The role of $\lambda_j$ and $a_g$ is not to change this ceiling, but to determine how closely each coordinate approaches it.

A more direct retention interpretation is available in the grouped normal-means representation. Under the Bayesian normal-means model $Y_j \mid \beta_j \sim \mathcal{N}(\beta_j, \sigma^2)$, conditional on the latent scales,

$$\mathbb{E}(\beta_j \mid Y_j, \lambda_j, a_g, \kappa_g, \tau, \sigma^2) = \frac{v_{j,g}}{\sigma^2 + v_{j,g}} Y_j.$$

Define the conditional retention factor

$$r_{j,g} := \frac{v_{j,g}}{\sigma^2 + v_{j,g}}. \tag{8}$$

Then $0 \leq r_{j,g} \leq 1$, and because $v_{j,g} \leq c_g^2$,

$$0 \leq r_{j,g} \leq \frac{c_g^2}{\sigma^2 + c_g^2} = \kappa_g. \tag{9}$$

Hence, in the grouped normal-means representation, $\kappa_g$ acts as a group-level upper bound on the conditional retention factor of every coordinate in group $g$.

This yields a two-stage interpretation of the hierarchy. At the group level, $\kappa_g$ determines the maximal fraction of raw signal that coordinates in group $g$ may retain after shrinkage; in this sense, $\kappa_g$ is the group-level allowance budget. Conditional on that budget, the local scales $\lambda_j$ and the group-amplitude factor $a_g$ determine how closely individual coordinates approach the common ceiling imposed by $\kappa_g$.

The grouped normal-means representation is introduced for analytical tractability and to make the shrinkage geometry explicit. The resulting coordinate-wise retention interpretation is exact in that representation, but should be viewed as a structural characterization of the hierarchy rather than a literal coordinate-wise identity for a general non-orthogonal regression design.

## 1.7 Prior on the allowance parameter

A convenient default, and the one adopted in the theoretical development below, is

$$\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \qquad \alpha_\kappa, \beta_\kappa > 0. \tag{10}$$

This choice is analytically tractable, flexible on $(0, 1)$, and particularly well suited to quantifying prior mass near the null regime $\kappa_g \approx 0$.

Under the one-to-one correspondence (2), the Beta prior on $\kappa_g$ induces a prior on the group-specific slab width $c_g$. One may therefore think either in terms of $\kappa_g$, which emphasizes group-level allowance, or in terms of $c_g$, which emphasizes group-specific slab regularization. These are mathematically equivalent parameterizations, but the $\kappa_g$ parameterization is more natural for the present allowance theory.

The roles of $\alpha_\kappa$ and $\beta_\kappa$ are distinct. The parameter $\alpha_\kappa$ controls how much prior mass accumulates near $\kappa_g = 0$, and therefore how strongly the prior favors aggressive group-level suppression. The parameter $\beta_\kappa$ shapes the upper tail on the unit interval and affects how willing the prior is to retain substantial group-level allowance.

## 1.8 Global-scale calibration

The prior for $\tau$ should reflect prior beliefs about overall sparsity. Following the regularized horseshoe literature, we recommend choosing the prior scale for $\tau$ through a sparsity-aware calibration rather than treating it as arbitrary.

Let $p_0$ denote a prior guess for the effective number of coefficients that are not strongly shrunk toward zero. Under standardized predictors, a natural baseline calibration is

$$\tilde{\tau}_0 \asymp \frac{p_0}{p - p_0} \frac{1}{\sqrt{n}}, \tag{11}$$

which then enters the conditional half-Cauchy prior through

$$\tau \mid \sigma \sim \mathrm{C}^+\big(0, \tilde{\tau}_0\, \sigma\big).$$

In the present grouped setting, we adopt this calibration principle as a practical default. The motivation is conceptual: $\tau$ should continue to encode *global sparsity*, while $\kappa_g$ should

encode *group-specific retention*. This division of labor keeps the hierarchy interpretable and aligns the grouped extension with the regularized horseshoe backbone. We do not claim here that (11) is an optimal calibration rule for the grouped hierarchy; rather, it serves as a principled default inherited from the regularized horseshoe backbone.

Thus, at the level of prior design, $\tau$ determines the global baseline shrinkage level, $\kappa_g$ redistributes retention capacity across groups, and $\lambda_j$ together with $a_g$ governs how retained capacity is allocated within groups.

## 1.9 Recommended working defaults

For practical implementation, we recommend the default specification

$$\begin{aligned} \lambda_j &\sim \mathrm{C}^+(0,1), \\ a_g &\sim \mathrm{HN}(0, s_a^2), \\ \kappa_g &\sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \\ \tau \mid \sigma &\sim \mathrm{C}^+\big(0, \tilde{\tau}_0\, \sigma\big), \end{aligned}$$

with

$$\tilde{\tau}_0 = \frac{p_0}{p - p_0} \frac{1}{\sqrt{n}},$$

where $p_0$ is a prior guess for the effective number of non-negligible coefficients.

For theory, the Beta family provides a convenient default for $\kappa_g$. In practice, $\mathrm{Beta}(1,1)$ is a simple baseline choice when little group-level prior information is available. The symmetric prior $\mathrm{Beta}(1/2, 1/2)$ is also a legitimate default, but it is U-shaped rather than flat: it places substantial prior mass near both $\kappa_g = 0$ and $\kappa_g = 1$. If stronger suppression of null groups is desired, one may choose a smaller $\alpha_\kappa$, while $\beta_\kappa$ continues to control how much prior mass remains near the upper end of the allowance range.

For the group-amplitude parameter $a_g$, a weakly informative half-normal scale $s_a$ is a reasonable default when no strong prior information is available about within-group heterogeneity. If a fully Bayesian treatment of the noise variance is desired, one may further augment the model with a prior $\sigma^2 \sim \Pi_\sigma$.

## 1.10 Posterior computation

Conditional on the latent scales, the model remains Gaussian in $\boldsymbol{\beta}$, so posterior computation inherits the main computational conveniences of global-local shrinkage priors. The hierarchy is only partially conjugate: exact named full conditional distributions are available for the Gaussian coefficient block and a subset of auxiliary variables, whereas several remaining one-dimensional updates are available only through exact conditional kernels. In particular, the coefficient block may be updated jointly under standard Gaussian formulas or sampled with modern gradient-based methods.

For computation it is often convenient to work with transformed variables such as

$$u_j := \lambda_j^2, \qquad b_g := a_g^2, \qquad \eta := \tau^2, \qquad q_g := c_g^2,$$

but these are deterministic reparameterizations of the primitive hierarchy rather than additional model components. For full Bayesian computation, it is numerically advantageous to work with transformed parameters such as

$$\log \tau, \qquad \log a_g, \qquad \log \lambda_j, \qquad \operatorname{logit}(\kappa_g),$$

which improve stability in weakly identified regions of the parameter space. In practice, we recommend implementing the full hierarchy through transformed, preferably non-centered, HMC/NUTS.

From a computational standpoint, the augmentation remains low-dimensional at the group level, introducing one allowance variable and one amplitude variable per group in addition to the standard regularized horseshoe components. Because $\tau$, $a_g$, and $\lambda_j$ enter multiplicatively in the effective variance, inference is usually most stable when substantive interpretation is concentrated on $\boldsymbol{\beta}$, the induced variances $v_{j,g}$, and the allowance parameters $\kappa_g$, rather than on any single latent scale in isolation.

The key point is structural: the present hierarchy admits an exact conditional decomposition, but several latent updates remain nonstandard. Accordingly, the computational representation should be read as a faithful decomposition of the primitive model rather than as a claim that the complete hierarchy is fully Gibbs-closed in elementary families.

### 1.11 Methodological summary

The grouped regularized horseshoe prior developed here is built around a simple organizing principle: each layer of the hierarchy is assigned a primary role in the prior construction.

| **Parameter** | **Primary role** |
|---|---|
| $\tau$ | global sparsity calibration |
| $\lambda_j$ | coordinate-specific local escape |
| $a_g$ | within-group amplitude modulation |
| $\kappa_g$ | group-specific allowance / retention gate |

The global and local layers coincide with those of the standard regularized horseshoe. The new ingredient is the allowance layer $\kappa_g$, which induces a group-specific slab width and serves as the main group-level control object in the present construction. These roles are prior-structural rather than fully orthogonal posterior roles, since the effective variance depends on the multiplicative combination $\tau a_g \lambda_j$.

## 2 Marginal prior theory for the grouped regularized horseshoe

This section studies the marginal prior behavior induced by the proposed grouped regularized horseshoe hierarchy. The main purpose is to clarify, in a precise analytical sense, how the new group-specific allowance layer modifies the usual regularized horseshoe backbone.

The results in this section are layered, and it is important to keep the three levels of marginalization separate.

First, conditional on both the combined global and group-amplitude scale $s_g := \tau a_g$ and the group-specific slab width $c_g$, the proposed prior retains the two signature features of the regularized horseshoe: a logarithmic singularity at the origin and finite-slab regularization in the tails.

Second, after the allowance layer is integrated out while $s_g$ is held fixed, the fixed-slab tail is replaced by a polynomial tail whose index is controlled by the upper-end Beta shape parameter $\beta_\kappa$.

Third, a further fully marginal tail or origin statement after integrating over $S = \tau a_g$ requires additional moment conditions and is therefore auxiliary rather than automatic under the default half-Cauchy / half-normal hierarchy. Accordingly, the fixed-$(s, c)$ results and the fixed-$s$ allowance-marginal results are the primary structural statements of this section; the fully marginal extensions later in the section should be read as conditional on extra integrability.

Throughout this section, $\sigma^2 > 0$ is treated as fixed. We focus on a single coordinate $j \in \mathcal{G}_g$ and suppress the indices when no confusion can arise. Write

$$s := s_g = \tau a_g > 0, \qquad c := c_g > 0,$$

and let $S := \tau a_g$ and $C := c_g$ denote the corresponding random variables under further marginalization. We also use

$$T := C^2$$

when it is more convenient to work with the slab variance parameter rather than the slab width itself.

We repeatedly use standard exponential-integral bounds of the form

$$\int_x^\infty \frac{e^{-t}}{t}\, dt = -\log x + O(1), \qquad x \downarrow 0,$$

and

$$\int_x^\infty \frac{e^{-t}}{t}\, dt \le K(1 + |\log x|), \qquad x > 0,$$

for a universal constant $K < \infty$.

Recall that

$$\beta \mid \lambda, s, c \sim \mathcal{N}(0, V), \qquad V = \frac{c^2 s^2 \lambda^2}{c^2 + s^2 \lambda^2}, \qquad \lambda \sim \mathrm{C}^+(0, 1). \tag{12}$$

Thus the conditional marginal prior of $\beta$ is a Gaussian scale mixture supported on $V \in (0, c^2)$.

## 2.1 Conditional marginal theory under fixed $s$ and $c$

All results in this subsection are *conditional* on fixed $(s, c) \in (0, \infty)^2$. They describe the regularized-horseshoe backbone of the grouped prior before any marginalization over the allowance layer.

**Proposition 2.1** (Induced variance density)**.** *Fix $s > 0$ and $c > 0$, and define*

$$V := \frac{c^2 s^2 \lambda^2}{c^2 + s^2 \lambda^2}, \qquad \lambda \sim \mathrm{C}^+(0,1).$$

*Then $V \in (0, c^2)$ almost surely, and $V$ has density*

$$f_V(v \mid s, c) = \frac{s\, c^3}{\pi \sqrt{v}\sqrt{c^2 - v}\,\left[s^2(c^2 - v) + c^2 v\right]}, \qquad 0 < v < c^2. \tag{13}$$

*Moreover,*

$$f_V(v \mid s, c) = \frac{1}{\pi s}\, v^{-1/2} + O(v^{1/2}), \qquad v \downarrow 0, \tag{14}$$

$$f_V(v \mid s, c) \sim \frac{s}{\pi c^2}\,(c^2 - v)^{-1/2}, \qquad v \uparrow c^2. \tag{15}$$

*In particular,*

$$f_V(v \mid s, c) \sim \frac{1}{\pi s}\, v^{-1/2}, \qquad v \downarrow 0. \tag{16}$$

*Proof.* Let

$$L := \lambda^2.$$

Since $\lambda \sim \mathrm{C}^+(0,1)$, the density of $L$ is

$$f_L(\ell) = \frac{1}{\pi\sqrt{\ell}(1+\ell)}, \qquad \ell > 0.$$

Now define

$$U := s^2 L.$$

By a change of variables,

$$f_U(u) = \frac{1}{s^2} f_L\!\left(\frac{u}{s^2}\right) = \frac{s}{\pi\sqrt{u}\,(s^2 + u)}, \qquad u > 0.$$

Next set

$$V = \frac{c^2 U}{c^2 + U}.$$

This map is strictly increasing from $(0, \infty)$ onto $(0, c^2)$, with inverse

$$U = \frac{c^2 V}{c^2 - V}, \qquad 0 < V < c^2,$$

and derivative

$$\frac{dU}{dV} = \frac{c^4}{(c^2 - V)^2}.$$

Therefore, by the change-of-variables formula,

$$f_V(v \mid s, c) = f_U\!\left(\frac{c^2 v}{c^2 - v}\right) \frac{c^4}{(c^2 - v)^2}, \qquad 0 < v < c^2.$$

Substituting the expression for $f_U$ gives

$$f_V(v \mid s,c) = \frac{s}{\pi} \frac{1}{\sqrt{\dfrac{c^2 v}{c^2 - v}} \left( s^2 + \dfrac{c^2 v}{c^2 - v} \right)} \cdot \frac{c^4}{(c^2 - v)^2}.$$

A straightforward simplification yields

$$f_V(v \mid s,c) = \frac{s\, c^3}{\pi \sqrt{v} \sqrt{c^2 - v}\, \left[ s^2 (c^2 - v) + c^2 v \right]}, \qquad 0 < v < c^2.$$

This proves (13).

It remains to verify the endpoint asymptotics. As $v \downarrow 0$, write

$$\sqrt{c^2 - v} = c\Big(1 - \frac{v}{c^2}\Big)^{1/2} = c\big(1 + O(v)\big),$$

and

$$s^2(c^2 - v) + c^2 v = s^2 c^2 + (c^2 - s^2) v = s^2 c^2 + O(v).$$

Hence

$$\frac{1}{\sqrt{c^2 - v}\, [s^2(c^2 - v) + c^2 v]} = \frac{1}{s^2 c^3} \big(1 + O(v)\big),$$

and therefore

$$f_V(v \mid s,c) = \frac{s\, c^3}{\pi}\, v^{-1/2} \cdot \frac{1}{s^2 c^3} \big(1 + O(v)\big) = \frac{1}{\pi s} v^{-1/2} \big(1 + O(v)\big).$$

This proves (14), and (16) follows immediately.

Similarly, as $v \uparrow c^2$,

$$\sqrt{v} \to c, \qquad s^2(c^2 - v) + c^2 v \to c^4,$$

so

$$f_V(v \mid s,c) \sim \frac{s\, c^3}{\pi c \sqrt{c^2 - v}\, c^4} = \frac{s}{\pi c^2} (c^2 - v)^{-1/2}.$$

This proves (15). □

As an immediate consequence, the conditional marginal prior density of $\beta$ admits the explicit integral representation

$$\pi(\beta \mid s,c) = \frac{s\, c^3}{\pi \sqrt{2\pi}} \int_0^{c^2} \frac{\exp(-\beta^2/2v)}{v \sqrt{c^2 - v}\, [s^2(c^2 - v) + c^2 v]}\, dv. \tag{17}$$

**Theorem 2.2** (Logarithmic singularity at the origin)**.** *Fix $s > 0$ and $c > 0$. Then the conditional marginal prior density $\pi(\beta \mid s,c)$ is unbounded at the origin and satisfies*

$$\pi(\beta \mid s,c) = \frac{2}{\pi s \sqrt{2\pi}} \log \frac{1}{|\beta|} + O(1), \qquad \beta \to 0. \tag{18}$$

*In particular,*

$$\pi(\beta \mid s,c) \to \infty \qquad \text{as } \beta \to 0.$$

*Proof.* Fix any $\delta \in (0, c^2)$. By Proposition 2.1, specifically (14),

$$f_V(v \mid s, c) = \frac{1}{\pi s} v^{-1/2} + R(v), \qquad 0 < v < \delta,$$

where

$$R(v) = O(v^{1/2}) \qquad (v \downarrow 0).$$

Hence

$$\pi(\beta \mid s, c) = \int_0^\delta \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} \left( \frac{1}{\pi s} v^{-1/2} + R(v) \right) dv + \int_\delta^{c^2} \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} f_V(v \mid s, c)\, dv.$$

First, the contribution from $[\delta, c^2)$ is uniformly bounded as $\beta \to 0$:

$$0 \le \int_\delta^{c^2} \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} f_V(v \mid s, c)\, dv \le \frac{1}{\sqrt{2\pi\delta}} \int_\delta^{c^2} f_V(v \mid s, c)\, dv \le \frac{1}{\sqrt{2\pi\delta}},$$

so this term is $O(1)$.

Second, since $R(v) = O(v^{1/2})$, there exists $M < \infty$ such that

$$|R(v)| \le M v^{1/2}, \qquad 0 < v < \delta.$$

Therefore

$$\left| \int_0^\delta \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} R(v)\, dv \right| \le \frac{M}{\sqrt{2\pi}} \int_0^\delta e^{-\beta^2/(2v)}\, dv \le \frac{M\delta}{\sqrt{2\pi}},$$

and hence this remainder contribution is also $O(1)$.

Thus the leading term is

$$\frac{1}{\pi s \sqrt{2\pi}} \int_0^\delta v^{-1} e^{-\beta^2/(2v)}\, dv.$$

With the substitution $t = \beta^2/(2v)$, so that $v = \beta^2/(2t)$ and $dv = -(\beta^2/2)t^{-2}dt$, we obtain

$$\int_0^\delta v^{-1} e^{-\beta^2/(2v)}\, dv = \int_{\beta^2/(2\delta)}^\infty \frac{e^{-t}}{t}\, dt.$$

Applying the standard exponential-integral expansion yields

$$\int_0^\delta v^{-1} e^{-\beta^2/(2v)}\, dv = 2 \log \frac{1}{|\beta|} + O(1), \qquad \beta \to 0.$$

Combining the preceding estimates,

$$\pi(\beta \mid s, c) = \frac{2}{\pi s \sqrt{2\pi}} \log \frac{1}{|\beta|} + O(1), \qquad \beta \to 0.$$

In particular, $\pi(\beta \mid s, c) \to \infty$ as $\beta \to 0$. □

**Proposition 2.3** (Finite-slab tail bound)**.** *Fix $s > 0$ and $c > 0$. Then for every $|\beta| > c$,*

$$\pi(\beta \mid s, c) \leq \frac{1}{\sqrt{2\pi}\, c} \exp\left(-\frac{\beta^2}{2c^2}\right). \tag{19}$$

*Proof.* Since $V \in (0, c^2)$ almost surely,

$$\pi(\beta \mid s, c) = \int_0^{c^2} \phi(\beta; 0, v)\, f_V(v \mid s, c)\, dv,$$

where

$$\phi(\beta; 0, v) = (2\pi v)^{-1/2} \exp\left(-\frac{\beta^2}{2v}\right).$$

For fixed $\beta \neq 0$, the derivative of $v \mapsto \phi(\beta; 0, v)$ is

$$\frac{\partial}{\partial v}\phi(\beta; 0, v) = \phi(\beta; 0, v)\, \frac{\beta^2 - v}{2v^2}.$$

Hence $v \mapsto \phi(\beta; 0, v)$ is increasing on $(0, \beta^2)$. If $|\beta| > c$, then $c^2 < \beta^2$, so for all $0 < v < c^2$,

$$\phi(\beta; 0, v) \leq \phi(\beta; 0, c^2) = \frac{1}{\sqrt{2\pi}\, c} \exp\left(-\frac{\beta^2}{2c^2}\right).$$

Integrating against the probability density $f_V(v \mid s, c)$ yields

$$\pi(\beta \mid s, c) \leq \frac{1}{\sqrt{2\pi}\, c} \exp\left(-\frac{\beta^2}{2c^2}\right),$$

as claimed. □

**Theorem 2.4** (Conditional tail asymptotics under a fixed allowance ceiling)**.** *Fix $s > 0$ and $c > 0$. Then*

$$\pi(\beta \mid s, c) \sim \frac{s}{\pi c\, |\beta|} \exp\left(-\frac{\beta^2}{2c^2}\right), \qquad |\beta| \to \infty. \tag{20}$$

*Proof.* Write $b := |\beta|$. The proof proceeds by showing that only a boundary layer near $v = c^2$ contributes at order $b^{-1}$. From (17),

$$\pi(\beta \mid s, c) = \frac{s\, c^3}{\pi\sqrt{2\pi}} \int_0^{c^2} \frac{\exp(-b^2/2v)}{v\sqrt{c^2 - v}\, [s^2(c^2 - v) + c^2 v]}\, dv. \tag{21}$$

Factor out the principal slab exponential:

$$\exp\left(-\frac{b^2}{2v}\right) = \exp\left(-\frac{b^2}{2c^2}\right) \exp\left(-\frac{b^2(c^2 - v)}{2c^2 v}\right).$$

Substituting this into (21) gives

$$\pi(\beta \mid s, c) = \exp\left(-\frac{b^2}{2c^2}\right) \frac{s\, c^3}{\pi\sqrt{2\pi}} \int_0^{c^2} \frac{\exp\left(-\frac{b^2(c^2 - v)}{2c^2 v}\right)}{v\sqrt{c^2 - v}\, [s^2(c^2 - v) + c^2 v]}\, dv. \tag{22}$$

Fix $\eta \in (0, c^2/2)$, and split the integral in (22) into

$$I_b^{\mathrm{far}} + I_b^{\mathrm{bd}},$$

where

$$I_b^{\mathrm{far}} := \int_0^{c^2-\eta} (\cdots)\, dv, \qquad I_b^{\mathrm{bd}} := \int_{c^2-\eta}^{c^2} (\cdots)\, dv.$$

We first show that the far contribution is negligible. Choose any fixed $\delta \in (0, c^2 - \eta)$, and decompose

$$I_b^{\mathrm{far}} = I_{b,1}^{\mathrm{far}} + I_{b,2}^{\mathrm{far}},$$

with

$$I_{b,1}^{\mathrm{far}} := \int_0^{\delta} \frac{\exp\left(-\dfrac{b^2(c^2-v)}{2c^2 v}\right)}{v\sqrt{c^2-v}\,[s^2(c^2-v)+c^2 v]}\, dv,$$

and

$$I_{b,2}^{\mathrm{far}} := \int_{\delta}^{c^2-\eta} \frac{\exp\left(-\dfrac{b^2(c^2-v)}{2c^2 v}\right)}{v\sqrt{c^2-v}\,[s^2(c^2-v)+c^2 v]}\, dv.$$

For $0 < v \le \delta$, we have

$$c^2 - v \ge c^2 - \delta, \qquad s^2(c^2-v) + c^2 v \ge s^2(c^2-v) \ge s^2(c^2-\delta),$$

and

$$\sqrt{c^2-v} \ge \sqrt{c^2-\delta}.$$

Therefore

$$0 \le I_{b,1}^{\mathrm{far}} \le \frac{1}{s^2(c^2-\delta)^{3/2}} \int_0^{\delta} v^{-1} \exp\left(-\frac{b^2(c^2-\delta)}{2c^2 v}\right)\, dv.$$

Set

$$a_\delta := \frac{c^2-\delta}{2c^2} > 0.$$

Then

$$I_{b,1}^{\mathrm{far}} \le \frac{1}{s^2(c^2-\delta)^{3/2}} \int_0^{\delta} v^{-1} e^{-a_\delta b^2/v}\, dv.$$

With the change of variables $t = a_\delta b^2/v$, this becomes

$$I_{b,1}^{\mathrm{far}} \le \frac{1}{s^2(c^2-\delta)^{3/2}} \int_{a_\delta b^2/\delta}^{\infty} \frac{e^{-t}}{t}\, dt.$$

Using

$$\int_x^{\infty} \frac{e^{-t}}{t}\, dt \le \frac{e^{-x}}{x}, \qquad x > 0,$$

we obtain

$$I_{b,1}^{\mathrm{far}} = O\left(b^{-2} e^{-a_\delta b^2/\delta}\right) = o(b^{-1}).$$

For $v \in [\delta, c^2 - \eta]$,

$$\frac{c^2 - v}{2c^2 v} \geq \frac{\eta}{2c^2(c^2 - \eta)} =: \rho_\eta > 0.$$

Also, on this interval,

$$v \geq \delta, \qquad \sqrt{c^2 - v} \geq \sqrt{\eta}, \qquad s^2(c^2 - v) + c^2 v \geq c^2 \delta.$$

Hence

$$0 \leq I_{b,2}^{\text{far}} \leq \frac{c^2 - \eta - \delta}{\delta \sqrt{\eta}\, c^2 \delta}\, e^{-\rho_\eta b^2} = O(e^{-\rho_\eta b^2}) = o(b^{-1}).$$

Thus

$$I_b^{\text{far}} = o(b^{-1}).$$

We next analyze the boundary contribution. Set

$$y := c^2 - v, \qquad 0 < y < \eta.$$

Then

$$I_b^{\text{bd}} = \int_0^\eta \frac{\exp\left(-\dfrac{b^2 y}{2c^2(c^2 - y)}\right)}{(c^2 - y)\sqrt{y}\,[s^2 y + c^2(c^2 - y)]}\, dy. \tag{23}$$

Now rescale the boundary layer by

$$y = \frac{2c^4}{b^2} u, \qquad dy = \frac{2c^4}{b^2}\, du.$$

Then $u \in (0, U_b)$, where

$$U_b := \frac{\eta b^2}{2c^4} \to \infty.$$

Substituting into (23), we obtain

$$I_b^{\text{bd}} = \frac{\sqrt{2}\, c^2}{b} \int_0^{U_b} H_b(u)\, du, \tag{24}$$

where

$$H_b(u) := \frac{\exp\left(-\dfrac{u}{1 - 2c^2 u/b^2}\right)}{\sqrt{u}\,(c^2 - 2c^4 u/b^2)\left[2s^2 c^4 u/b^2 + c^2(c^2 - 2c^4 u/b^2)\right]}.$$

Extend $H_b$ by zero outside $(0, U_b)$. Then for each fixed $u > 0$,

$$H_b(u)\mathbf{1}_{(0,U_b)}(u) \longrightarrow \frac{e^{-u}}{c^6\sqrt{u}} \qquad (b \to \infty).$$

Moreover, for all sufficiently large $b$ and all $u > 0$,

$$0 \leq H_b(u)\mathbf{1}_{(0,U_b)}(u) \leq \frac{4}{c^6} u^{-1/2} e^{-u}.$$

Indeed, on $(0, U_b)$ we have $2c^2u/b^2 < \eta/c^2 < 1/2$, so

$$c^2 - \frac{2c^4u}{b^2} \geq \frac{c^2}{2},$$

and

$$2s^2c^4u/b^2 + c^2(c^2 - 2c^4u/b^2) \geq \frac{c^4}{2},$$

while

$$\exp\left(-\frac{u}{1 - 2c^2u/b^2}\right) \leq e^{-u}.$$

Since $u^{-1/2}e^{-u} \in L^1(0, \infty)$, dominated convergence yields

$$\int_0^{U_b} H_b(u)\, du = \int_0^\infty H_b(u)\mathbf{1}_{(0,U_b)}(u)\, du \longrightarrow \frac{1}{c^6}\int_0^\infty u^{-1/2}e^{-u}\, du = \frac{\sqrt{\pi}}{c^6}.$$

Hence, from (24),

$$I_b^{\mathrm{bd}} \sim \frac{\sqrt{2}\, c^2}{b} \cdot \frac{\sqrt{\pi}}{c^6} = \frac{\sqrt{2\pi}}{c^4\, b}.$$

Combining $I_b^{\mathrm{far}} = o(b^{-1})$ with the preceding asymptotic for $I_b^{\mathrm{bd}}$, we obtain from (22)

$$\pi(\beta \mid s, c) = \exp\left(-\frac{b^2}{2c^2}\right) \frac{s\, c^3}{\pi\sqrt{2\pi}} \left(\frac{\sqrt{2\pi}}{c^4\, b} + o(b^{-1})\right).$$

Therefore

$$\pi(\beta \mid s, c) \sim \frac{s}{\pi c\, b} \exp\left(-\frac{b^2}{2c^2}\right).$$

Since $b = |\beta|$, this is exactly (20). □

**Remark 2.5.** *The proof of Theorem 2.4 shows that the $|\beta|^{-1}$ prefactor is driven by the square-root singularity of the induced variance density at the slab ceiling $c^2$. Thus the large-$|\beta|$ behavior is controlled by the boundary layer $v \uparrow c^2$, while the interior region contributes only exponentially negligible mass relative to the leading term.*

## 2.2 Allowance-induced slab-width distribution

We now move from the fixed-$(s, c)$ analysis to the first genuinely new group-level ingredient, namely the distribution induced on the slab parameter by the allowance prior. In this subsection, the key object is the group-specific slab variance $T = C^2$, obtained by pushing forward the Beta prior on $\kappa_g$.

**Proposition 2.6** (Induced prior on the group-specific slab variance parameter)**.** *Suppose*

$$\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \qquad \alpha_\kappa, \beta_\kappa > 0,$$

*and define*

$$T := c_g^2 = \frac{\sigma^2\kappa_g}{1 - \kappa_g}.$$

*Then $T \in (0,\infty)$ has density*

$$f_T(t) = \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa,\beta_\kappa)}\, t^{\alpha_\kappa - 1}(\sigma^2+t)^{-(\alpha_\kappa+\beta_\kappa)}, \qquad t>0. \tag{25}$$

*In particular,*

$$f_T(t) \sim \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa,\beta_\kappa)}\, t^{-(\beta_\kappa+1)}, \qquad t\to\infty. \tag{26}$$

*Proof.* The transformation

$$\kappa = \frac{t}{\sigma^2+t}, \qquad t>0,$$

is a bijection from $(0,\infty)$ onto $(0,1)$, with inverse

$$t = \frac{\sigma^2\kappa}{1-\kappa},$$

and derivative

$$\frac{d\kappa}{dt} = \frac{\sigma^2}{(\sigma^2+t)^2}.$$

Since $\kappa_g \sim \mathrm{Beta}(\alpha_\kappa,\beta_\kappa)$, its density is

$$f_\kappa(\kappa) = \frac{1}{B(\alpha_\kappa,\beta_\kappa)}\kappa^{\alpha_\kappa-1}(1-\kappa)^{\beta_\kappa-1}, \qquad 0<\kappa<1.$$

Applying the change-of-variables formula,

$$f_T(t) = f_\kappa\left(\frac{t}{\sigma^2+t}\right)\frac{\sigma^2}{(\sigma^2+t)^2}.$$

Substituting the Beta density gives

$$f_T(t) = \frac{1}{B(\alpha_\kappa,\beta_\kappa)}\left(\frac{t}{\sigma^2+t}\right)^{\alpha_\kappa-1}\left(\frac{\sigma^2}{\sigma^2+t}\right)^{\beta_\kappa-1}\frac{\sigma^2}{(\sigma^2+t)^2}.$$

Collecting powers yields

$$f_T(t) = \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa,\beta_\kappa)}\, t^{\alpha_\kappa-1}(\sigma^2+t)^{-(\alpha_\kappa+\beta_\kappa)}, \qquad t>0.$$

This proves (25).

Finally, as $t\to\infty$,

$$(\sigma^2+t)^{-(\alpha_\kappa+\beta_\kappa)} = t^{-(\alpha_\kappa+\beta_\kappa)}(1+o(1)),$$

so

$$f_T(t) \sim \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa,\beta_\kappa)}\, t^{\alpha_\kappa-1-(\alpha_\kappa+\beta_\kappa)} = \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa,\beta_\kappa)}\, t^{-(\beta_\kappa+1)}.$$

This proves (26). □

**Corollary 2.7** (Scaled Beta-prime characterization and moment formulas)**.** *Under the assumptions of Proposition 2.6, define*

$$W := \frac{T}{\sigma^2} = \frac{\kappa_g}{1-\kappa_g}.$$

*Then $W$ has the Beta-prime distribution with shape parameters $(\alpha_\kappa, \beta_\kappa)$, that is,*

$$f_W(w) = \frac{1}{B(\alpha_\kappa, \beta_\kappa)} w^{\alpha_\kappa - 1}(1+w)^{-(\alpha_\kappa+\beta_\kappa)}, \qquad w > 0. \tag{27}$$

*Moreover, for every real $m$ such that*

$$-\alpha_\kappa < m < \beta_\kappa,$$

*the $m$-th moment of $T$ exists and is given by*

$$E[T^m] = (\sigma^2)^m \frac{B(\alpha_\kappa + m, \beta_\kappa - m)}{B(\alpha_\kappa, \beta_\kappa)}. \tag{28}$$

*Equivalently,*

$$E[T^m] < \infty \qquad \Longleftrightarrow \qquad -\alpha_\kappa < m < \beta_\kappa. \tag{29}$$

*If $C := c_g = \sqrt{T}$, then for every real $r$ such that*

$$-2\alpha_\kappa < r < 2\beta_\kappa,$$

*the $r$-th moment of $C$ exists and satisfies*

$$E[C^r] = \sigma^r \frac{B(\alpha_\kappa + r/2, \beta_\kappa - r/2)}{B(\alpha_\kappa, \beta_\kappa)}. \tag{30}$$

*In particular,*

$$E[C^{-1}] < \infty \qquad \Longleftrightarrow \qquad \alpha_\kappa > \frac{1}{2}.$$

*Proof.* Since $T = \sigma^2 W$, the density of $W$ follows immediately from (25) by the change of variables $t = \sigma^2 w$, $dt = \sigma^2\, dw$:

$$f_W(w) = f_T(\sigma^2 w)\, \sigma^2 = \frac{1}{B(\alpha_\kappa, \beta_\kappa)} w^{\alpha_\kappa - 1}(1+w)^{-(\alpha_\kappa+\beta_\kappa)}, \qquad w > 0.$$

Thus $W \sim \mathrm{BetaPrime}(\alpha_\kappa, \beta_\kappa)$.

For the moment formula, let $m \in \mathbb{R}$. Then

$$E[T^m] = (\sigma^2)^m E[W^m] = (\sigma^2)^m \frac{1}{B(\alpha_\kappa, \beta_\kappa)} \int_0^\infty w^{\alpha_\kappa + m - 1}(1+w)^{-(\alpha_\kappa+\beta_\kappa)}\, dw.$$

The integral is finite exactly when

$$\alpha_\kappa + m > 0 \qquad \text{and} \qquad \beta_\kappa - m > 0,$$

that is, when $-\alpha_\kappa < m < \beta_\kappa$. In that case,

$$\int_0^\infty w^{\alpha_\kappa + m - 1}(1+w)^{-(\alpha_\kappa+\beta_\kappa)}\, dw = B(\alpha_\kappa + m, \beta_\kappa - m),$$

which proves (28) and (29).

Finally, since $C = \sqrt{T}$, we have $C^r = T^{r/2}$. Applying (28) with $m = r/2$ gives (30). The last statement follows by setting $r = -1$. □

## 2.3 Tail behavior after integrating over the allowance layer

Throughout this subsection, $s > 0$ is treated as fixed, and the allowance-induced slab parameter $T = c_g^2$ is integrated with respect to its marginal law (25). Thus the results below are still *conditional on the combined scale* $s = \tau a_g$. They describe the structural effect of integrating over the allowance layer itself, and should not be read as fully marginal statements under the default half-Cauchy / half-normal hierarchy.

We use the convention that

$$\pi(\beta \mid s, t)$$

denotes the conditional density $\pi(\beta \mid s, c)$ evaluated at $c = \sqrt{t}$. Define

$$\pi(\beta \mid s) := \int_0^\infty \pi(\beta \mid s, t)\, f_T(t)\, dt,$$

where $f_T$ is the density in (25).

**Theorem 2.8** (Allowance-induced polynomial tail under fixed $s$)**.** *Suppose* $\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa)$ *with* $\alpha_\kappa, \beta_\kappa > 0$*, and let* $T = c_g^2$ *as above. Then for every fixed* $s > 0$*,*

$$\pi(\beta \mid s) \sim C_\kappa(s, \sigma^2)\, |\beta|^{-(2\beta_\kappa+2)}, \qquad |\beta| \to \infty, \tag{31}$$

*where*

$$C_\kappa(s, \sigma^2) = \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa, \beta_\kappa)} \cdot \frac{s\, 2^{\beta_\kappa+1/2}\Gamma(\beta_\kappa+1)\Gamma(\beta_\kappa)}{\pi\, \Gamma(\beta_\kappa+1/2)}. \tag{32}$$

*In particular,* $\pi(\beta \mid s)$ *has regularly varying tails with index*

$$-(2\beta_\kappa + 2).$$

*Proof.* Let $x := \beta$ and $b := |x|$. From (17), with $c^2 = t$, we have

$$\pi(x \mid s, t) = \frac{s\, t^{3/2}}{\pi\sqrt{2\pi}} \int_0^t \frac{\exp(-x^2/2v)}{v\,\sqrt{t-v}\,[s^2(t-v)+tv]}\, dv.$$

With the substitution $v = tw$, $0 < w < 1$, this becomes

$$\pi(x \mid s, t) = \frac{s}{\pi\sqrt{2\pi}} \int_0^1 \frac{\exp(-x^2/(2tw))}{w\sqrt{1-w}\,[s^2(1-w)+tw]}\, dw. \tag{33}$$

Integrating (33) against $f_T(t)\, dt$, Tonelli's theorem applies because the integrand is nonnegative. Thus

$$\pi(x \mid s) = \frac{s\,\sigma^{2\beta_\kappa}}{\pi\sqrt{2\pi}\, B(\alpha_\kappa, \beta_\kappa)} \int_0^\infty \int_0^1 \frac{\exp(-x^2/(2tw))\, t^{\alpha_\kappa-1}(\sigma^2+t)^{-(\alpha_\kappa+\beta_\kappa)}}{w\sqrt{1-w}\,[s^2(1-w)+tw]}\, dw\, dt. \tag{34}$$

Now set $t = b^2 z$, so $dt = b^2 dz$. Then

$$\pi(x \mid s) = \frac{s\,\sigma^{2\beta_\kappa}}{\pi\sqrt{2\pi}\, B(\alpha_\kappa, \beta_\kappa)}\, b^{-(2\beta_\kappa+2)} \int_0^\infty \int_0^1 G_b(z, w)\, dw\, dz,$$

where

$$G_b(z,w) := \frac{\exp(-1/(2zw))\, z^{\alpha_\kappa - 1}(z+\sigma^2/b^2)^{-(\alpha_\kappa+\beta_\kappa)}}{w\sqrt{1-w}\,[zw + s^2(1-w)/b^2]}.$$

Fix $z > 0$ and $w \in (0,1)$. As $b \to \infty$,

$$z + \sigma^2/b^2 \to z, \qquad zw + s^2(1-w)/b^2 \to zw,$$

hence

$$G_b(z,w) \to \exp(-1/(2zw)) z^{-(\beta_\kappa+2)} w^{-2}(1-w)^{-1/2} =: G(z,w).$$

We now verify domination. Since $z + \sigma^2/b^2 \geq z$,

$$(z+\sigma^2/b^2)^{-(\alpha_\kappa+\beta_\kappa)} \leq z^{-(\alpha_\kappa+\beta_\kappa)}.$$

Therefore

$$z^{\alpha_\kappa-1}(z+\sigma^2/b^2)^{-(\alpha_\kappa+\beta_\kappa)} \leq z^{\alpha_\kappa-1} z^{-(\alpha_\kappa+\beta_\kappa)} = z^{-(\beta_\kappa+1)}.$$

Also,

$$zw + s^2(1-w)/b^2 \geq zw.$$

Hence

$$0 \leq G_b(z,w) \leq \exp(-1/(2zw))\, z^{-(\beta_\kappa+2)} w^{-2}(1-w)^{-1/2} = G(z,w).$$

It remains to show that $G \in L^1((0,\infty)\times(0,1))$. For fixed $w \in (0,1)$,

$$\int_0^\infty e^{-1/(2zw)} z^{-(\beta_\kappa+2)}\, dz = (2w)^{\beta_\kappa+1}\Gamma(\beta_\kappa+1),$$

which follows from the substitution $u = 1/(2zw)$. Therefore

$$\int_0^\infty G(z,w)\, dz = 2^{\beta_\kappa+1}\Gamma(\beta_\kappa+1) w^{\beta_\kappa-1}(1-w)^{-1/2}.$$

Since $\beta_\kappa > 0$, the function

$$w \mapsto w^{\beta_\kappa-1}(1-w)^{-1/2}$$

is integrable on $(0,1)$. Hence $G \in L^1((0,\infty)\times(0,1))$, and dominated convergence gives

$$\int_0^\infty \int_0^1 G_b(z,w)\, dw\, dz \longrightarrow \int_0^\infty \int_0^1 G(z,w)\, dw\, dz.$$

We now compute the limit constant:

$$\begin{aligned}
\int_0^\infty \int_0^1 G(z,w)\, dw\, dz &= 2^{\beta_\kappa+1}\Gamma(\beta_\kappa+1) \int_0^1 w^{\beta_\kappa-1}(1-w)^{-1/2}\, dw \\
&= 2^{\beta_\kappa+1}\Gamma(\beta_\kappa+1) B(\beta_\kappa, 1/2) \\
&= 2^{\beta_\kappa+1}\Gamma(\beta_\kappa+1)\Gamma(\beta_\kappa)\frac{\sqrt{\pi}}{\Gamma(\beta_\kappa+1/2)}.
\end{aligned}$$

Substituting this back yields

$$\pi(\beta \mid s) \sim \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa,\beta_\kappa)} \cdot \frac{s\, 2^{\beta_\kappa+1/2}\Gamma(\beta_\kappa+1)\Gamma(\beta_\kappa)}{\pi\,\Gamma(\beta_\kappa+1/2)}\, |\beta|^{-(2\beta_\kappa+2)}.$$

This proves (31) and (32). □

**Remark 2.9.** *Theorem 2.8 is a fixed-s result and therefore applies directly to the allowance-marginalized hierarchy conditional on the combined scale $S = \tau a_g$. It is the primary tail characterization for the allowance layer itself. A further fully marginal tail statement after integrating over $S$ requires additional moment conditions and is not automatic under the default half-Cauchy / half-normal hierarchy.*

**Remark 2.10.** *A notable consequence of Theorem 2.8 is that the eventual marginal tail index is determined entirely by the upper-end Beta shape parameter $\beta_\kappa$. By contrast, $\alpha_\kappa$ influences prior mass near $\kappa_g = 0$ but does not affect the asymptotic tail exponent.*

## 2.4 One-coordinate reduction to the regularized horseshoe

**Proposition 2.11** (One-coordinate reduction)**.** *Suppose $p_g = 1$ and $a_g \equiv 1$. Then*

$$V = \frac{c_g^2 \tau^2 \lambda_j^2}{c_g^2 + \tau^2 \lambda_j^2},$$

*which is exactly the effective variance of the standard regularized horseshoe with slab width $c_g$. Consequently, all conditional marginal results above reduce to the corresponding one-coordinate regularized horseshoe statements with group-specific slab width $c_g$.*

*Proof.* If $p_g = 1$ and $a_g \equiv 1$, then the conditional variance in (12) becomes

$$V = \frac{c_g^2 \tau^2 \lambda_j^2}{c_g^2 + \tau^2 \lambda_j^2}.$$

This is exactly the effective variance appearing in the standard regularized horseshoe with slab width $c_g$. Therefore the induced one-coordinate prior coincides with the regularized horseshoe prior having slab parameter $c_g$, and every conditional marginal statement proved above specializes immediately to the corresponding regularized-horseshoe statement. □

**Remark 2.12.** *The proposed grouped hierarchy is not merely analogous to the regularized horseshoe; it restricts exactly to it under the one-coordinate specialization $p_g = 1$ together with $a_g \equiv 1$.*

## 2.5 Further marginalization over $S = \tau a_g$

We now consider a further layer of marginalization over the combined scale $S := \tau a_g$. The results in this subsection are *auxiliary extensions* under additional moment control on $S$ (and occasionally on $C$). They are useful for clarifying how the conditional structure propagates through further marginalization, but they should not be read as automatic consequences of the default half-Cauchy / half-normal hierarchy.

### 2.5.1 Uniform origin expansion on compact rectangles

**Proposition 2.13** (Uniform logarithmic expansion on compact rectangles)**.** *Fix a compact rectangle*

$$K := [s_1, s_2] \times [c_1, c_2] \subset (0, \infty)^2, \qquad 0 < s_1 \le s_2 < \infty, \quad 0 < c_1 \le c_2 < \infty.$$

*Then there exists a constant $M_K < \infty$ such that for all $(s,c) \in K$ and all $0 < |\beta| < 1$,*

$$\left|\pi(\beta \mid s,c) - \frac{2}{\pi s\sqrt{2\pi}} \log \frac{1}{|\beta|}\right| \le M_K. \tag{35}$$

*Proof.* Fix

$$\delta_K := \frac{c_1^2}{2}.$$

For every $(s,c) \in K$, we have $\delta_K < c^2$, so we may decompose

$$\pi(\beta \mid s,c) = I_1(\beta; s,c) + I_2(\beta; s,c),$$

where

$$I_1(\beta; s,c) := \int_0^{\delta_K} \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} f_V(v \mid s,c)\, dv,$$

and

$$I_2(\beta; s,c) := \int_{\delta_K}^{c^2} \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} f_V(v \mid s,c)\, dv.$$

Write

$$f_V(v \mid s,c) = \frac{1}{\pi} v^{-1/2} A(v; s,c),$$

where

$$A(v; s,c) := \frac{s\, c^3}{\sqrt{c^2 - v}\,[s^2(c^2 - v) + c^2 v]}.$$

Since $A$ is smooth on the compact set $[0, \delta_K] \times K$, its derivative with respect to $v$ is bounded there. Also,

$$A(0; s,c) = \frac{1}{s}.$$

Hence, by the mean value theorem, there exists $L_K < \infty$ such that

$$\left|A(v; s,c) - \frac{1}{s}\right| \le L_K v, \qquad 0 \le v \le \delta_K, \quad (s,c) \in K.$$

Therefore

$$f_V(v \mid s,c) = \frac{1}{\pi s} v^{-1/2} + R_K(v; s,c),$$

where

$$|R_K(v; s,c)| \le \frac{L_K}{\pi} v^{1/2}, \qquad 0 \le v \le \delta_K, \quad (s,c) \in K.$$

Substituting into $I_1$ gives

$$I_1(\beta; s,c) = \frac{1}{\pi s\sqrt{2\pi}} \int_0^{\delta_K} v^{-1} e^{-\beta^2/(2v)}\, dv + E_1(\beta; s,c),$$

with

$$|E_1(\beta; s,c)| \le \frac{L_K}{\pi\sqrt{2\pi}} \int_0^{\delta_K} e^{-\beta^2/(2v)}\, dv \le \frac{L_K \delta_K}{\pi\sqrt{2\pi}} =: M_{K,1}.$$

Moreover,

$$\int_0^{\delta_K} v^{-1} e^{-\beta^2/(2v)}\, dv = \int_{\beta^2/(2\delta_K)}^{\infty} \frac{e^{-t}}{t}\, dt = 2\log\frac{1}{|\beta|} + O_K(1),$$

uniformly for $0 < |\beta| < 1$, because $\delta_K$ is fixed. Thus

$$I_1(\beta; s, c) = \frac{2}{\pi s\sqrt{2\pi}} \log\frac{1}{|\beta|} + O_K(1), \tag{36}$$

uniformly over $(s, c) \in K$.

We next bound $I_2$. For $v \in [\delta_K, c^2)$ and $(s, c) \in K$,

$$v \geq \delta_K, \qquad s^2(c^2 - v) + c^2 v \geq c^2 v \geq c_1^2 \delta_K, \qquad s\, c^3 \leq s_2 c_2^3.$$

Hence, by (13),

$$f_V(v \mid s, c) \leq \frac{s_2 c_2^3}{\pi\sqrt{\delta_K}\sqrt{c^2 - v}\, c_1^2 \delta_K} \leq \frac{M_{K,2}}{\sqrt{c^2 - v}}$$

for some finite constant $M_{K,2}$. Therefore

$$0 \leq I_2(\beta; s, c) \leq \frac{M_{K,2}}{\sqrt{2\pi\delta_K}} \int_{\delta_K}^{c^2} \frac{dv}{\sqrt{c^2 - v}} \leq \frac{M_{K,2}}{\sqrt{2\pi\delta_K}} \int_0^{c_2^2} \frac{dy}{\sqrt{y}} =: M_{K,3},$$

uniformly in $(s, c) \in K$ and $0 < |\beta| < 1$.

Combining this with (36), we obtain

$$\left| \pi(\beta \mid s, c) - \frac{2}{\pi s\sqrt{2\pi}} \log\frac{1}{|\beta|} \right| \leq M_K$$

for some finite constant $M_K$, uniformly over $(s, c) \in K$ and $0 < |\beta| < 1$. This proves (35). □

**Proposition 2.14** (Persistence of the origin spike)**.** *Let $(S, C)$ be a positive random vector, and define the fully marginal prior density*

$$\bar{\pi}(\beta) := \iint_{(0,\infty)^2} \pi(\beta \mid s, c)\, dF_{S,C}(s, c).$$

*Then*

$$\bar{\pi}(\beta) \to \infty \qquad \text{as } \beta \to 0.$$

*Proof.* Since $(S, C)$ takes values in $(0, \infty)^2$ almost surely, and

$$(0, \infty)^2 = \bigcup_{m=1}^{\infty} [m^{-1}, m] \times [m^{-1}, m],$$

there exists $m_0 \in \mathbb{N}$ such that

$$\mathbb{P}\big((S, C) \in K\big) > 0, \qquad K := [m_0^{-1}, m_0] \times [m_0^{-1}, m_0].$$

By Proposition 2.13, there exists $M_K < \infty$ such that for all $(s, c) \in K$ and $0 < |\beta| < 1$,

$$\pi(\beta \mid s, c) \geq \frac{2}{\pi s\sqrt{2\pi}} \log \frac{1}{|\beta|} - M_K \geq \frac{2}{\pi m_0\sqrt{2\pi}} \log \frac{1}{|\beta|} - M_K.$$

Hence

$$\begin{aligned}
\bar{\pi}(\beta) &= \iint_{(0,\infty)^2} \pi(\beta \mid s, c)\, dF_{S,C}(s, c) \\
&\geq \iint_K \pi(\beta \mid s, c)\, dF_{S,C}(s, c) \\
&\geq \mathbb{P}\big((S, C) \in K\big) \left[\frac{2}{\pi m_0\sqrt{2\pi}} \log \frac{1}{|\beta|} - M_K\right].
\end{aligned}$$

Since $\mathbb{P}((S, C) \in K) > 0$, the right-hand side diverges to $+\infty$ as $\beta \to 0$. Therefore

$$\bar{\pi}(\beta) \to \infty \qquad (\beta \to 0).$$

□

### 2.5.2 Fully marginal tail behavior under a finite first moment

Recall from Theorem 2.8 that for fixed $s > 0$,

$$\pi(\beta \mid s) \sim C_\kappa(s, \sigma^2)\, |\beta|^{-(2\beta_\kappa+2)},$$

with

$$C_\kappa(s, \sigma^2) = K_\kappa(\sigma^2)\, s,$$

where

$$K_\kappa(\sigma^2) := \frac{\sigma^{2\beta_\kappa}}{B(\alpha_\kappa, \beta_\kappa)} \cdot \frac{2^{\beta_\kappa+1/2}\Gamma(\beta_\kappa + 1)\Gamma(\beta_\kappa)}{\pi\, \Gamma(\beta_\kappa + 1/2)}. \tag{37}$$

**Theorem 2.15** (Preservation of the polynomial tail after integrating over $S$)**.** *Let $S$ be a positive random variable, independent of the allowance layer, and define*

$$\bar{\pi}(\beta) := \int_0^\infty \pi(\beta \mid s)\, dF_S(s),$$

*where $\pi(\beta \mid s)$ denotes the allowance-marginalized density from Theorem 2.8. If*

$$E[S] < \infty,$$

*then*

$$\bar{\pi}(\beta) \sim K_\kappa(\sigma^2)\, E[S]\, |\beta|^{-(2\beta_\kappa+2)}, \qquad |\beta| \to \infty, \tag{38}$$

*In particular, the fully marginal prior retains the same tail index*

$$-(2\beta_\kappa + 2)$$

*as in Theorem 2.8.*

*Proof.* Let $x := \beta$ and $b := |x|$. From the proof of Theorem 2.8, we have the exact representation

$$\pi(x \mid s) = K_0\, s\, b^{-(2\beta_\kappa+2)} \int_0^\infty \int_0^1 G_b(z, w; s)\, dw\, dz,$$

where

$$K_0 := \frac{\sigma^{2\beta_\kappa}}{\pi\sqrt{2\pi}\, B(\alpha_\kappa, \beta_\kappa)},$$

and

$$G_b(z, w; s) := \frac{\exp(-1/(2zw))\, z^{\alpha_\kappa-1}(z + \sigma^2/b^2)^{-(\alpha_\kappa+\beta_\kappa)}}{w\sqrt{1-w}\,[zw + s^2(1-w)/b^2]}.$$

As in the proof of Theorem 2.8,

$$0 \le G_b(z, w; s) \le G(z, w),$$

where

$$G(z, w) := \exp(-1/(2zw)) z^{-(\beta_\kappa+2)} w^{-2}(1-w)^{-1/2},$$

and $G \in L^1((0,\infty) \times (0,1))$. Therefore

$$\int_0^\infty \int_0^1 G_b(z, w; s)\, dw\, dz \le \int_0^\infty \int_0^1 G(z, w)\, dw\, dz =: M_\kappa < \infty,$$

uniformly in $s > 0$ and $b \ge 1$. Consequently,

$$0 \le \pi(\beta \mid s) \le K_0 M_\kappa\, s\, |\beta|^{-(2\beta_\kappa+2)}, \qquad |\beta| \ge 1. \tag{39}$$

On the other hand, for each fixed $s > 0$,

$$\frac{\pi(\beta \mid s)}{|\beta|^{-(2\beta_\kappa+2)}} \longrightarrow K_\kappa(\sigma^2)\, s, \qquad |\beta| \to \infty,$$

by Theorem 2.8. Dividing (39) by $|\beta|^{-(2\beta_\kappa+2)}$, we obtain the integrable bound

$$0 \le \frac{\pi(\beta \mid s)}{|\beta|^{-(2\beta_\kappa+2)}} \le K_0 M_\kappa\, s.$$

Since $E[S] < \infty$, dominated convergence gives

$$\frac{\bar{\pi}(\beta)}{|\beta|^{-(2\beta_\kappa+2)}} = \int_0^\infty \frac{\pi(\beta \mid s)}{|\beta|^{-(2\beta_\kappa+2)}}\, dF_S(s) \longrightarrow \int_0^\infty K_\kappa(\sigma^2)\, s\, dF_S(s).$$

Therefore

$$\bar{\pi}(\beta) \sim K_\kappa(\sigma^2)\, E[S]\, |\beta|^{-(2\beta_\kappa+2)}, \qquad |\beta| \to \infty,$$

which proves (38). □

**Remark 2.16.** *Theorem 2.15 is an auxiliary fully marginal extension under the moment condition $E[S] < \infty$. It does not automatically apply under the default half-Cauchy / half-normal specification, since that default hierarchy does not satisfy the required moment condition. Accordingly, for the default hierarchy, the fixed-s tail characterization in Theorem 2.8 remains the primary structural result proved here.*

**Remark 2.17.** *Within the present marginal theory, the amplitude layer enters the leading tail constant through the combined scale $S = \tau a_g$, whereas the allowance layer controls the eventual tail exponent after marginalization.*

**Remark 2.18.** *Under the default independent hierarchy*

$$\tau \mid \sigma \sim \mathrm{C}^{+}\big(0, \tilde{\tau}_0\, \sigma\big), \qquad a_g \sim \mathrm{HN}(0, s_a^2),$$

*one has $E[S] = \infty$ and $E[S^{-1}] = \infty$ for $S = \tau a_g$. Thus the fully marginal tail and origin theorems below should not be read as automatic properties of the default specification.*

**Remark 2.19.** *The moment assumptions used below to obtain fully marginal origin asymptotics are sufficient conditions that justify an integrable domination argument. They are not claimed to be sharp, and need not hold under the default hierarchy.*

**Theorem 2.20** (Fully marginal logarithmic singularity under an integrable inverse scale)**.** *Let $(S, C)$ be a positive random vector, and define the fully marginal prior density*

$$\bar{\pi}(\beta) := \iint_{(0,\infty)^2} \pi(\beta \mid s, c)\, dF_{S,C}(s, c).$$

*Assume that*

$$E\big[S^{-1}\big(1 + |\log C|\big)\big] < \infty, \qquad E[C^{-1}] < \infty. \tag{40}$$

*Then*

$$\frac{\bar{\pi}(\beta)}{\log(1/|\beta|)} \longrightarrow \frac{2}{\pi\sqrt{2\pi}}\, E[S^{-1}], \qquad \beta \to 0. \tag{41}$$

*Equivalently,*

$$\bar{\pi}(\beta) \sim \frac{2}{\pi\sqrt{2\pi}}\, E[S^{-1}] \log \frac{1}{|\beta|}, \qquad \beta \to 0. \tag{42}$$

*In particular, the fully marginal prior remains unbounded at the origin.*

*Proof.* Let

$$L_\beta := \log \frac{1}{|\beta|}, \qquad 0 < |\beta| < e^{-1},$$

so that $L_\beta \geq 1$. Define

$$H_\beta(s, c) := \frac{\pi(\beta \mid s, c)}{L_\beta}.$$

For each fixed $s > 0$ and $c > 0$, Theorem 2.2 yields

$$\pi(\beta \mid s, c) = \frac{2}{\pi s\sqrt{2\pi}} L_\beta + O_{s,c}(1), \qquad \beta \to 0,$$

and hence

$$H_\beta(s, c) \longrightarrow \frac{2}{\pi s\sqrt{2\pi}}, \qquad \beta \to 0.$$

Thus it remains to produce an integrable dominating function.

Decompose

$$\pi(\beta \mid s, c) = I_1(\beta; s, c) + I_2(\beta; s, c),$$

where

$$I_1(\beta; s, c) := \int_0^{c^2/2} \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} f_V(v \mid s, c)\, dv,$$

and

$$I_2(\beta; s, c) := \int_{c^2/2}^{c^2} \frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} f_V(v \mid s, c)\, dv.$$

We first bound $I_1$. For $0 < v \le c^2/2$, write $r = v/c^2 \in (0, 1/2]$. By (13),

$$f_V(v \mid s, c) = \frac{1}{\pi} v^{-1/2} \frac{s}{\sqrt{1-r}\,[s^2(1-r) + c^2 r]}.$$

Since $r \le 1/2$,

$$\sqrt{1-r} \ge 2^{-1/2}, \qquad s^2(1-r) + c^2 r \ge \frac{s^2}{2}.$$

Therefore

$$f_V(v \mid s, c) \le \frac{2\sqrt{2}}{\pi s}\, v^{-1/2}, \qquad 0 < v \le c^2/2.$$

It follows that

$$\begin{aligned} I_1(\beta; s, c) &\le \frac{2\sqrt{2}}{\pi s \sqrt{2\pi}} \int_0^{c^2/2} v^{-1} e^{-\beta^2/(2v)}\, dv \\ &= \frac{2\sqrt{2}}{\pi s \sqrt{2\pi}} \int_{\beta^2/c^2}^{\infty} \frac{e^{-t}}{t}\, dt. \end{aligned}$$

Applying the standard bound

$$\int_x^{\infty} \frac{e^{-t}}{t}\, dt \le K(1 + |\log x|), \qquad x > 0,$$

we get

$$I_1(\beta; s, c) \le \frac{K_1}{s} \Big(1 + \big|2 \log|\beta| - \log(c^2)\big|\Big)$$

for some finite constant $K_1$. Since $L_\beta = \log(1/|\beta|) \ge 1$,

$$\frac{I_1(\beta; s, c)}{L_\beta} \le K_2\, s^{-1}(1 + |\log c|)$$

for some constant $K_2 < \infty$, uniformly in $0 < |\beta| < e^{-1}$.

We next bound $I_2$. On the interval $[c^2/2, c^2]$, we have $v \ge c^2/2$, so

$$\frac{1}{\sqrt{2\pi v}} e^{-\beta^2/(2v)} \le \frac{1}{\sqrt{\pi}\, c}.$$

Since $f_V(\cdot \mid s, c)$ is a probability density,

$$I_2(\beta; s, c) \le \frac{1}{\sqrt{\pi}\, c} \int_{c^2/2}^{c^2} f_V(v \mid s, c)\, dv \le \frac{1}{\sqrt{\pi}\, c}.$$

Hence

$$\frac{I_2(\beta; s, c)}{L_\beta} \le \frac{1}{\sqrt{\pi}\, c}.$$

Combining the two estimates, we obtain

$$0 \le H_\beta(s, c) \le K_2\, s^{-1}(1 + |\log c|) + \frac{1}{\sqrt{\pi}\, c} =: D(s, c),$$

uniformly for $0 < |\beta| < e^{-1}$. By assumption (40), $D$ is integrable with respect to $F_{S,C}$. Therefore dominated convergence applies:

$$\begin{aligned} \frac{\bar{\pi}(\beta)}{L_\beta} &= \iint H_\beta(s, c)\, dF_{S,C}(s, c) \\ &\longrightarrow \iint \frac{2}{\pi s \sqrt{2\pi}}\, dF_{S,C}(s, c) \\ &= \frac{2}{\pi\sqrt{2\pi}} E[S^{-1}]. \end{aligned}$$

This proves (41), and hence (42). In particular, $\bar{\pi}(\beta) \to \infty$ as $\beta \to 0$. □

**Remark 2.21.** *Theorem 2.20 is likewise an auxiliary extension under the integrability assumptions in* (40)*. It does not automatically apply under the default half-Cauchy / half-normal hierarchy, for which $E[S^{-1}] = \infty$. Accordingly, for the default hierarchy, the conditional-on-$(s, c)$ origin characterization in Theorem 2.2 remains the primary structural description proved here.*

**Remark 2.22** (Sufficient conditions under an independent allowance prior)**.** *Suppose $S$ and $C$ are independent, and $C^2$ is induced by the Beta allowance prior in Proposition 2.6. Then*

$$E[|\log C|] < \infty$$

*for all $\alpha_\kappa, \beta_\kappa > 0$, and*

$$E[C^{-1}] < \infty \qquad \Longleftrightarrow \qquad \alpha_\kappa > \frac{1}{2}.$$

*Hence, under independence, a sufficient condition for Theorem 2.20 is*

$$E[S^{-1}] < \infty \qquad \text{and} \qquad \alpha_\kappa > \frac{1}{2}.$$

*Under the default independent hierarchy $\tau \mid \sigma \sim \mathrm{C}^+\big(0, \tilde{\tau}_0\, \sigma\big)$ and $a_g \sim \mathrm{HN}(0, s_a^2)$, one has $E[S^{-1}] = \infty$, where $S = \tau a_g$. Accordingly, the conditional-on-$(s, c)$ origin characterization remains the more natural structural description near zero for the default hierarchy.*

## 2.6 Interpretation

The marginal prior theory developed above leads to a layered structural summary.

First, conditional on $s = \tau a_g$ and $c_g$, the grouped prior preserves the two defining regularized horseshoe features: a logarithmic spike at zero and finite-slab regularization in the tails. This is the direct grouped analogue of the ordinary regularized horseshoe backbone.

Second, once the allowance layer is integrated out while $s$ is held fixed, the tail regime changes. The fixed-slab exponential decay is replaced by a polynomial tail whose index is controlled exactly by the Beta upper-end parameter $\beta_\kappa$. In this sense, $\kappa_g$, through the induced slab width $c_g$, determines how much tail protection is available to the coordinates in group $g$.

Third, the fully marginal origin and tail statements obtained after further integrating over $S = \tau a_g$ are auxiliary extensions under additional moment conditions. They clarify how the conditional structure can propagate through further marginalization when suitable integrability holds, but they are not automatic properties of the default half-Cauchy / half-normal hierarchy.

Finally, when $p_g = 1$ and $a_g \equiv 1$, the construction reduces exactly to the ordinary regularized horseshoe.

Thus the proposed hierarchy may be viewed as a grouped regularized horseshoe with a group-specific allowance mechanism: the regularized horseshoe backbone is retained, and the new allowance layer randomizes the slab ceiling at the group level in a way that leaves the conditional origin singularity order intact while converting the fixed-slab tail into a Beta-controlled polynomial tail after marginalization over the allowance layer.

# 3 Posterior theory for the allowance parameter

This section develops posterior theory for the group-specific allowance parameter $\kappa_g$. The results are organized in two layers.

First, under the full grouped normal-means hierarchy, we establish exact structural results for the posterior of $\kappa_g$, including an exact bridge from posterior allowance to posterior mean control, posterior propriety of the conditional $\kappa_g$-posterior, and a conditional posterior geometry given the latent variance ratios.

Second, to obtain sharp asymptotic posterior concentration results, we study a *profile specialization* of the full hierarchy, obtained by fixing

$$\lambda_j \equiv 1, \qquad a_g \equiv 1, \qquad \tau > 0 \text{ fixed.}$$

Under this specialization, the posterior analysis of $\kappa_g$ becomes one-dimensional, while the role of $\kappa_g$ as the group-level allowance parameter remains fully explicit. The resulting theory yields null-group posterior contraction, signal-side localization and retention, simultaneous groupwise separation under fixed $G$, and a direct lift to blockwise orthonormal grouped regression.

Throughout this section, $\sigma^2 > 0$ is treated as fixed. In the sharp asymptotic results below, $\tau > 0$ is also treated as a fixed deterministic constant and is therefore omitted from conditioning notation whenever no confusion can arise. The exact structural results

in the first subsection hold within the grouped normal-means representation. They provide posterior insight into the role of $\kappa_g$, but should not be read as literal coordinate-wise identities for a general non-orthogonal regression design.

## 3.1 Full hierarchy: exact bridge and conditional posterior geometry

We work under the grouped normal-means representation

$$Y_j \mid \beta_j \sim \mathcal{N}(\beta_j, \sigma^2), \qquad j = 1, \ldots, p, \tag{43}$$

with groups

$$\{1, \ldots, p\} = \mathcal{G}_1 \sqcup \cdots \sqcup \mathcal{G}_G, \qquad |\mathcal{G}_g| = p_g.$$

Recall that under the full grouped regularized horseshoe hierarchy,

$$\beta_j \mid \lambda_j, a_g, \kappa_g, \tau, \sigma^2 \sim \mathcal{N}(0, v_{j,g}),$$

with

$$v_{j,g} = \frac{\sigma^2 \kappa_g \tau^2 \lambda_j^2 a_g^2}{\sigma^2 \kappa_g + (1-\kappa_g)\tau^2 \lambda_j^2 a_g^2} \le \frac{\sigma^2 \kappa_g}{1-\kappa_g}.$$

**Theorem 3.1** (Exact bridge from posterior allowance to posterior mean)**.** *Fix a group $g$. Under* (43) *and the full grouped regularized horseshoe hierarchy, define*

$$\omega_j(\boldsymbol{Y}, \sigma^2) := \mathbb{E}\left( \frac{v_{j,g}}{\sigma^2 + v_{j,g}} \,\middle|\, \boldsymbol{Y}, \sigma^2 \right), \qquad j \in \mathcal{G}_g.$$

*Then*

$$\mathbb{E}(\beta_j \mid \boldsymbol{Y}, \sigma^2) = \omega_j(\boldsymbol{Y}, \sigma^2)\, Y_j, \qquad j \in \mathcal{G}_g, \tag{44}$$

*and*

$$0 \le \omega_j(\boldsymbol{Y}, \sigma^2) \le \mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2) \qquad a.s. \tag{45}$$

*Consequently,*

$$\|\mathbb{E}(\boldsymbol{\beta}_g \mid \boldsymbol{Y}, \sigma^2)\|_2 \le \mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2)\, \|\boldsymbol{Y}_g\|_2 \qquad a.s. \tag{46}$$

*Proof.* Conditional on all hyperparameters,

$$\mathbb{E}(\beta_j \mid Y_j, \lambda_j, a_g, \kappa_g, \tau, \sigma^2) = \frac{v_{j,g}}{\sigma^2 + v_{j,g}}\, Y_j.$$

Let

$$c_g^2 := \frac{\sigma^2 \kappa_g}{1-\kappa_g}.$$

Since $v_{j,g} \le c_g^2$ and

$$\frac{c_g^2}{\sigma^2 + c_g^2} = \kappa_g,$$

we have

$$0 \le \frac{v_{j,g}}{\sigma^2 + v_{j,g}} \le \kappa_g.$$

Taking posterior expectation over the hyperparameters given $(\boldsymbol{Y}, \sigma^2)$ yields (44) and (45). Then

$$|\mathbb{E}(\beta_j \mid \boldsymbol{Y}, \sigma^2)| \le \mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2)\, |Y_j|.$$

Squaring, summing over $j \in \mathcal{G}_g$, and taking square roots yields (46). □

To study the conditional posterior of $\kappa_g$, it is convenient to condition on the latent variance ratios

$$\chi_j := \frac{\tau^2 \lambda_j^2 a_g^2}{\sigma^2}, \qquad j \in \mathcal{G}_g. \tag{47}$$

Then

$$v_{j,g} = \sigma^2 \frac{\kappa_g \chi_j}{\kappa_g + (1-\kappa_g)\chi_j},$$

and the marginal variance of $Y_j$ given $(\kappa_g, \boldsymbol{\chi}_g)$ is

$$\sigma^2 m_j(\kappa_g), \qquad m_j(\kappa) := \frac{\kappa + \chi_j}{\kappa + (1-\kappa)\chi_j}.$$

Write

$$t_j := \frac{Y_j^2}{\sigma^2}, \qquad \mathbf{t}_g := (t_j : j \in \mathcal{G}_g), \qquad \boldsymbol{\chi}_g := (\chi_j : j \in \mathcal{G}_g).$$

Assuming

$$\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \qquad \alpha_\kappa, \beta_\kappa > 0,$$

the conditional posterior kernel of $\kappa_g$ given $(\mathbf{t}_g, \boldsymbol{\chi}_g)$ is

$$q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) \propto \kappa^{\alpha_\kappa - 1}(1-\kappa)^{\beta_\kappa - 1} \prod_{j \in \mathcal{G}_g} m_j(\kappa)^{-1/2} \exp\left(-\frac{t_j}{2\, m_j(\kappa)}\right), \qquad 0 < \kappa < 1. \tag{48}$$

**Proposition 3.2** (Posterior propriety of the conditional $\kappa_g$-posterior)**.** *Fix $\boldsymbol{\chi}_g \in (0,\infty)^{p_g}$ and suppose $\alpha_\kappa, \beta_\kappa > 0$. Then the kernel in (48) is integrable over $(0,1)$, and hence defines a proper conditional posterior density for $\kappa_g$.*

*Proof.* For each $j \in \mathcal{G}_g$, the map $\kappa \mapsto m_j(\kappa)$ is continuous and strictly positive on the compact interval $[0,1]$. Hence there exist finite constants $0 < m_{j,-} \le m_{j,+} < \infty$ such that

$$m_{j,-} \le m_j(\kappa) \le m_{j,+}, \qquad 0 \le \kappa \le 1.$$

Therefore

$$C_1 \le \prod_{j \in \mathcal{G}_g} m_j(\kappa)^{-1/2} \exp\left(-\frac{t_j}{2\, m_j(\kappa)}\right) \le C_2, \qquad 0 < \kappa < 1,$$

for finite constants $0 < C_1 \le C_2 < \infty$ depending on $(\mathbf{t}_g, \boldsymbol{\chi}_g)$, but not on $\kappa$. Hence the kernel (48) is comparable to

$$\kappa^{\alpha_\kappa - 1}(1-\kappa)^{\beta_\kappa - 1},$$

which is integrable on $(0,1)$ because $\alpha_\kappa, \beta_\kappa > 0$. Thus the conditional posterior is proper. □

Define further

$$z_j(\kappa) := \frac{1}{m_j(\kappa)} = \frac{\kappa + (1-\kappa)\chi_j}{\kappa + \chi_j}, \qquad h_j(\kappa) := 1 - z_j(\kappa) = \frac{\kappa\chi_j}{\kappa + \chi_j}. \tag{49}$$

Then

$$z_j'(\kappa) = -\frac{\chi_j^2}{(\kappa+\chi_j)^2} < 0, \qquad h_j'(\kappa) = \frac{\chi_j^2}{(\kappa+\chi_j)^2} > 0. \tag{50}$$

**Theorem 3.3** (Coordinatewise data monotonicity)**.** *Suppose* $\mathbf{t}_g' = (t_j')_{j\in\mathcal{G}_g}$ *and* $\mathbf{t}_g = (t_j)_{j\in\mathcal{G}_g}$ *satisfy*

$$t_j' \geq t_j, \qquad j \in \mathcal{G}_g.$$

*Then*

$$\kappa \longmapsto \frac{q_g(\kappa \mid \mathbf{t}_g', \boldsymbol{\chi}_g)}{q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)}$$

*is nondecreasing on* $(0,1)$*. Consequently, the normalized conditional posterior distributions satisfy*

$$\kappa_g \mid \mathbf{t}_g', \boldsymbol{\chi}_g \quad \textit{stochastically dominates} \quad \kappa_g \mid \mathbf{t}_g, \boldsymbol{\chi}_g.$$

*In particular, for every bounded increasing measurable* $\varphi : (0,1) \to \mathbb{R}$*,*

$$\mathbb{E}[\varphi(\kappa_g) \mid \mathbf{t}_g', \boldsymbol{\chi}_g] \geq \mathbb{E}[\varphi(\kappa_g) \mid \mathbf{t}_g, \boldsymbol{\chi}_g].$$

*Proof.* From (48),

$$\frac{q_g(\kappa \mid \mathbf{t}_g', \boldsymbol{\chi}_g)}{q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)} \propto \exp\left(-\frac{1}{2}\sum_{j\in\mathcal{G}_g}(t_j' - t_j)z_j(\kappa)\right).$$

Since $t_j' - t_j \geq 0$ and $z_j(\kappa)$ is strictly decreasing by (50), the exponent above is nondecreasing in $\kappa$. Hence the density ratio is nondecreasing on $(0,1)$, which implies first-order stochastic dominance and the expectation inequality. □

**Remark 3.4** (Reduction by equal-$\chi$ classes)**.** *If* $\chi_j$ *takes only finitely many distinct values* $v_1, \ldots, v_L$*, then the conditional posterior depends on the data only through the classwise sums*

$$T_\ell := \sum_{j:\chi_j = v_\ell} t_j, \qquad \ell = 1, \ldots, L.$$

*The homogeneous case* $\chi_j \equiv \chi_g$ *corresponds to* $L = 1$*, in which case the conditional posterior depends on the data only through*

$$T_g = \sum_{j\in\mathcal{G}_g} t_j.$$

**Proposition 3.5** (Exact posterior score identity)**.** *Let* $\ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)$ *denote any log-kernel of* $q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)$*. Then, for* $0 < \kappa < 1$*,*

$$\frac{d}{d\kappa}\ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) = \frac{\alpha_\kappa - 1}{\kappa} - \frac{\beta_\kappa - 1}{1-\kappa} + \frac{1}{2}\sum_{j\in\mathcal{G}_g}\frac{\chi_j^2}{(\kappa+\chi_j)^2}\big(t_j - m_j(\kappa)\big). \tag{51}$$

*Proof.* From (48),

$$\ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) = (\alpha_\kappa - 1)\log\kappa + (\beta_\kappa - 1)\log(1-\kappa) - \frac{1}{2}\sum_{j\in\mathcal{G}_g}\log m_j(\kappa) - \frac{1}{2}\sum_{j\in\mathcal{G}_g} t_j z_j(\kappa) + C,$$

where $C$ does not depend on $\kappa$. Since

$$\frac{d}{d\kappa}\log m_j(\kappa) = -\frac{z_j'(\kappa)}{z_j(\kappa)},$$

we obtain

$$\frac{d}{d\kappa}\ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) = \frac{\alpha_\kappa - 1}{\kappa} - \frac{\beta_\kappa - 1}{1-\kappa} + \frac{1}{2}\sum_{j\in\mathcal{G}_g} z_j'(\kappa)\left(\frac{1}{z_j(\kappa)} - t_j\right).$$

Now use $\frac{1}{z_j(\kappa)} = m_j(\kappa)$ and $z_j'(\kappa) = -\chi_j^2/(\kappa+\chi_j)^2$, which yields (51). □

**Corollary 3.6** (Finite-part derivative at $\kappa = 0$)**.** *Define*

$$\ell_{g,0}^*(\kappa) := \ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) - (\alpha_\kappa - 1)\log\kappa.$$

*Then $\ell_{g,0}^*$ admits a finite right derivative at 0, given by*

$$(\ell_{g,0}^*)'(0+) = -(\beta_\kappa - 1) + \frac{1}{2}\sum_{j\in\mathcal{G}_g}(t_j - 1). \tag{52}$$

*Proof.* Subtract $(\alpha_\kappa - 1)/\kappa$ from (51) and let $\kappa \downarrow 0$. Since $m_j(0) = 1$ and $\chi_j^2/(\kappa+\chi_j)^2 \to 1$, the limit is (52). □

**Corollary 3.7** (Finite-part derivative at $\kappa = 1$)**.** *Define*

$$\ell_{g,1}^*(\kappa) := \ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) - (\beta_\kappa - 1)\log(1-\kappa).$$

*Then $\ell_{g,1}^*$ admits a finite left derivative at 1, given by*

$$(\ell_{g,1}^*)'(1-) = (\alpha_\kappa - 1) + \frac{1}{2}\sum_{j\in\mathcal{G}_g}\frac{\chi_j^2}{(1+\chi_j)^2}\big(t_j - (1+\chi_j)\big). \tag{53}$$

*Proof.* Add $(\beta_\kappa - 1)/(1-\kappa)$ to (51) and let $\kappa \uparrow 1$. Since $m_j(1) = 1+\chi_j$ and $\chi_j^2/(\kappa+\chi_j)^2 \to \chi_j^2/(1+\chi_j)^2$, the limit is (53). □

**Remark 3.8** (Boundary interpretation)**.** *Corollaries 3.6 and 3.7 quantify the local directional tendency after removing the Beta-induced endpoint singularity. In particular, when $\alpha_\kappa = 1$, the local behavior at $\kappa = 0$ is determined by the sign of* (52)*; when $\beta_\kappa = 1$, the local behavior at $\kappa = 1$ is determined by the sign of* (53)*.*

**Proposition 3.9** (Exact likelihood-ratio decomposition relative to an arbitrary baseline)**.** *Define the conditional likelihood kernel*

$$L_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) \propto \prod_{j \in \mathcal{G}_g} m_j(\kappa)^{-1/2} \exp\left(-\frac{t_j}{2\, m_j(\kappa)}\right).$$

*Then, for any* $\kappa, \kappa_0 \in (0,1)$*,*

$$\log \frac{L_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)}{L_g(\kappa_0 \mid \mathbf{t}_g, \boldsymbol{\chi}_g)} = \frac{1}{2} \sum_{j \in \mathcal{G}_g} \left[\log \frac{1 - h_j(\kappa)}{1 - h_j(\kappa_0)} + t_j\big(h_j(\kappa) - h_j(\kappa_0)\big)\right]. \tag{54}$$

*Proof.* Because $1 - h_j(\kappa) = z_j(\kappa) = 1/m_j(\kappa)$, we have

$$-\log m_j(\kappa) = \log(1 - h_j(\kappa)), \qquad \frac{1}{m_j(\kappa)} = 1 - h_j(\kappa).$$

Hence

$$\begin{aligned}\log L_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) &= -\frac{1}{2} \sum_{j \in \mathcal{G}_g} \log m_j(\kappa) - \frac{1}{2} \sum_{j \in \mathcal{G}_g} t_j \frac{1}{m_j(\kappa)} + C \\ &= \frac{1}{2} \sum_{j \in \mathcal{G}_g} \log(1 - h_j(\kappa)) - \frac{1}{2} \sum_{j \in \mathcal{G}_g} t_j\big(1 - h_j(\kappa)\big) + C,\end{aligned}$$

where $C$ does not depend on $\kappa$. Subtracting the same expression at $\kappa_0$ yields (54). □

**Proposition 3.10** (Exact posterior-kernel ratio decomposition)**.** *For any* $\kappa, \kappa_0 \in (0,1)$*,*

$$\begin{aligned}\log \frac{q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)}{q_g(\kappa_0 \mid \mathbf{t}_g, \boldsymbol{\chi}_g)} &= (\alpha_\kappa - 1) \log \frac{\kappa}{\kappa_0} + (\beta_\kappa - 1) \log \frac{1 - \kappa}{1 - \kappa_0} \\ &\quad + \frac{1}{2} \sum_{j \in \mathcal{G}_g} \left[\log \frac{1 - h_j(\kappa)}{1 - h_j(\kappa_0)} + t_j\big(h_j(\kappa) - h_j(\kappa_0)\big)\right].\end{aligned} \tag{55}$$

*Proof.* This follows by adding the prior log-ratio

$$(\alpha_\kappa - 1) \log(\kappa/\kappa_0) + (\beta_\kappa - 1) \log\big((1 - \kappa)/(1 - \kappa_0)\big)$$

to the likelihood-ratio identity (54). □

**Remark 3.11** (Signal–complexity interpretation)**.** *In* (54)*, the term*

$$\frac{1}{2} \sum_{j \in \mathcal{G}_g} \log \frac{1 - h_j(\kappa)}{1 - h_j(\kappa_0)}$$

*acts as a complexity adjustment, whereas*

$$\frac{1}{2} \sum_{j \in \mathcal{G}_g} t_j\big(h_j(\kappa) - h_j(\kappa_0)\big)$$

*acts as a signal reward. Thus moving from* $\kappa_0$ *to* $\kappa$ *is favored by the data precisely when the signal reward outweighs the complexity penalty.*

**Theorem 3.12** (Interior local mode from boundary score directions)**.** *Let*

$$\ell_g(\kappa) := \ell_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g).$$

*Suppose there exist* $\varepsilon_0, \varepsilon_1 > 0$ *such that*

$$\ell_g'(\kappa) > 0, \qquad 0 < \kappa < \varepsilon_0,$$

*and*

$$\ell_g'(\kappa) < 0, \qquad 1 - \varepsilon_1 < \kappa < 1.$$

*Then* $q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)$ *admits at least one local maximizer in* $(0, 1)$*.*

*Proof.* Since $q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)$ is continuous and strictly positive on any compact subinterval of $(0, 1)$, it attains a maximum on $[a, b]$ for any $0 < a < b < 1$. Choose $a \in (0, \varepsilon_0)$ and $b \in (1 - \varepsilon_1, 1)$. By hypothesis, $q_g$ is strictly increasing on $(0, a)$ and strictly decreasing on $(b, 1)$. Therefore a maximizer of $q_g$ on $[a, b]$ is automatically a local maximizer in $(0, 1)$. □

**Corollary 3.13** (A verifiable sufficient condition for an interior local mode)**.** *The boundary score conditions in Theorem 3.12 hold if the following left-boundary condition and right-boundary condition are both satisfied.*

***Left boundary:*** *either*

$$\alpha_\kappa > 1,$$

*or*

$$\alpha_\kappa = 1 \quad \textit{and} \quad -(\beta_\kappa - 1) + \frac{1}{2} \sum_{j \in \mathcal{G}_g} (t_j - 1) > 0.$$

***Right boundary:*** *either*

$$\beta_\kappa > 1,$$

*or*

$$\beta_\kappa = 1 \quad \textit{and} \quad (\alpha_\kappa - 1) + \frac{1}{2} \sum_{j \in \mathcal{G}_g} \frac{\chi_j^2}{(1 + \chi_j)^2} \big(t_j - (1 + \chi_j)\big) < 0.$$

*Under these conditions,* $q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g)$ *admits at least one interior local maximizer in* $(0, 1)$*.*

*Proof.* If $\alpha_\kappa > 1$, then the term $(\alpha_\kappa - 1)/\kappa$ in (51) dominates as $\kappa \downarrow 0$, so $\ell_g'(\kappa) > 0$ for all sufficiently small $\kappa > 0$. If $\alpha_\kappa = 1$, the claim follows from Corollary 3.6.

Similarly, if $\beta_\kappa > 1$, then the term $-(\beta_\kappa - 1)/(1 - \kappa)$ in (51) dominates as $\kappa \uparrow 1$, so $\ell_g'(\kappa) < 0$ for all sufficiently large $\kappa < 1$. If $\beta_\kappa = 1$, the claim follows from Corollary 3.7.

Thus the hypotheses of Theorem 3.12 are satisfied. □

**Proposition 3.14** (Homogeneous-group reduction)**.** *Suppose that within group* $g$*,*

$$\chi_j \equiv \chi_g > 0, \qquad j \in \mathcal{G}_g.$$

*Define the group energy statistic*

$$T_g := \sum_{j\in\mathcal{G}_g} t_j = \sum_{j\in\mathcal{G}_g} \frac{Y_j^2}{\sigma^2}. \tag{56}$$

*Then the conditional posterior kernel of $\kappa$ depends on the data only through $T_g$, and may be written as*

$$q_g(\kappa \mid T_g, \chi_g) \propto \kappa^{\alpha_\kappa - 1}(1-\kappa)^{\beta_\kappa - 1} m_g(\kappa)^{-p_g/2} \exp\left(-\frac{T_g}{2\, m_g(\kappa)}\right), \qquad 0 < \kappa < 1, \tag{57}$$

*where*

$$m_g(\kappa) := \frac{\kappa + \chi_g}{\kappa + (1-\kappa)\chi_g}. \tag{58}$$

*Equivalently, the posterior score reduces to*

$$\ell_g'(\kappa) = \frac{\alpha_\kappa - 1}{\kappa} - \frac{\beta_\kappa - 1}{1-\kappa} + \frac{1}{2}\frac{\chi_g^2}{(\kappa+\chi_g)^2}\big(T_g - p_g\, m_g(\kappa)\big). \tag{59}$$

*Proof.* If $\chi_j \equiv \chi_g$ for all $j \in \mathcal{G}_g$, then

$$m_j(\kappa) \equiv m_g(\kappa) = \frac{\kappa + \chi_g}{\kappa + (1-\kappa)\chi_g}.$$

Substituting this into (48) yields

$$q_g(\kappa \mid \mathbf{t}_g, \boldsymbol{\chi}_g) \propto \kappa^{\alpha_\kappa - 1}(1-\kappa)^{\beta_\kappa - 1} m_g(\kappa)^{-p_g/2} \exp\left(-\frac{1}{2m_g(\kappa)} \sum_{j\in\mathcal{G}_g} t_j\right),$$

which is exactly (57). Hence the dependence on the data is only through $T_g = \sum_{j\in\mathcal{G}_g} t_j$.

Similarly, (51) becomes

$$\ell_g'(\kappa) = \frac{\alpha_\kappa - 1}{\kappa} - \frac{\beta_\kappa - 1}{1-\kappa} + \frac{1}{2}\sum_{j\in\mathcal{G}_g} \frac{\chi_g^2}{(\kappa+\chi_g)^2}\big(t_j - m_g(\kappa)\big),$$

which simplifies to (59). □

**Corollary 3.15** (Group-energy threshold under a flat allowance prior)**.** *Assume the homogeneous setting of Proposition 3.14 and take the flat allowance prior*

$$\alpha_\kappa = \beta_\kappa = 1.$$

*Then*

$$(\ell_{g,0}^*)'(0+) = \frac{1}{2}\big(T_g - p_g\big), \tag{60}$$

*and*

$$(\ell_{g,1}^*)'(1-) = \frac{1}{2}\frac{\chi_g^2}{(1+\chi_g)^2}\big(T_g - p_g(1+\chi_g)\big). \tag{61}$$

*Consequently, if*

$$p_g < T_g < p_g(1+\chi_g), \tag{62}$$

*then the conditional posterior kernel $q_g(\kappa \mid T_g, \chi_g)$ admits at least one interior local maximizer in $(0,1)$.*

*Proof.* Under $\alpha_\kappa = \beta_\kappa = 1$, the left-boundary finite-part formula (52) becomes

$$(\ell^*_{g,0})'(0+) = \frac{1}{2}\sum_{j\in\mathcal{G}_g}(t_j - 1) = \frac{1}{2}(T_g - p_g),$$

which proves (60).

Likewise, the right-boundary finite-part formula (53) becomes

$$(\ell^*_{g,1})'(1-) = \frac{1}{2}\sum_{j\in\mathcal{G}_g}\frac{\chi_g^2}{(1+\chi_g)^2}\big(t_j - (1+\chi_g)\big) = \frac{1}{2}\frac{\chi_g^2}{(1+\chi_g)^2}\big(T_g - p_g(1+\chi_g)\big),$$

which proves (61).

If (62) holds, then

$$(\ell^*_{g,0})'(0+) > 0 \qquad \text{and} \qquad (\ell^*_{g,1})'(1-) < 0.$$

Therefore the hypotheses of Corollary 3.13 are satisfied, and the posterior kernel admits at least one interior local maximizer. □

## 3.2 Profile specialization and one-dimensional posterior profiles

We now turn to sharp asymptotic posterior theory under the *profile specialization*

$$\lambda_j \equiv 1, \qquad a_g \equiv 1, \qquad \tau > 0 \text{ fixed.} \tag{63}$$

All asymptotic statements in Subsections 3.3– 3.5 are understood along a sequence of grouped problems indexed by $n \to \infty$. For a given groupwise sequence, we suppress the index $n$ and write $p_g, \boldsymbol{\beta}_{0,g}, \boldsymbol{Y}_g, \kappa_g$ instead of $p_{g,n}, \boldsymbol{\beta}_{0,g,n}, \boldsymbol{Y}_{g,n}, \kappa_{g,n}$. Thus, whenever we write $p_g \to \infty$, $\mu_g/p_g \to \xi$, or similar asymptotic relations, these are understood along the underlying sequence.

Under (63), $\tau$ is a fixed constant rather than a random parameter. Accordingly, the posterior statements below concern the exact one-dimensional posterior of each allowance parameter $\kappa_g$ under this specialization.

In the profile specialization, the effective prior variance is common within each group:

$$v(\kappa) := \frac{\sigma^2\kappa\tau^2}{\sigma^2\kappa + (1-\kappa)\tau^2}, \qquad \rho := \tau/\sigma > 0. \tag{64}$$

Define

$$\theta(\kappa) := \frac{v(\kappa)}{\sigma^2} = \frac{\kappa\rho^2}{\kappa + (1-\kappa)\rho^2}, \tag{65}$$

$$\psi(\kappa) := \frac{v(\kappa)}{\sigma^2 + v(\kappa)} = \frac{\kappa\rho^2}{\rho^2 + \kappa}, \tag{66}$$

$$\phi(x) := -\log(1+x) + \frac{x}{1+x}, \qquad x \geq 0. \tag{67}$$

Then $\phi(0) = 0$, $\phi(x) < 0$ for $x > 0$, and

$$\phi(x) = -\frac{x^2}{2} + O(x^3) \qquad \text{as } x \downarrow 0.$$

For group $g$, let

$$S_g := \frac{\|\boldsymbol{Y}_g\|_2^2}{2\sigma^2}, \qquad \mu_g := \frac{\|\boldsymbol{\beta}_{0,g}\|_2^2}{2\sigma^2}, \qquad W_g := S_g - \frac{p_g}{2} - \mu_g. \tag{68}$$

The profile log Bayes factor against $\kappa = 0$ is

$$\ell_g(\kappa) = -\frac{p_g}{2}\log(1 + \theta(\kappa)) + S_g\psi(\kappa). \tag{69}$$

Since

$$S_g = \frac{p_g}{2} + \mu_g + W_g,$$

this may be rewritten as

$$\ell_g(\kappa) = \frac{p_g}{2}\phi(\theta(\kappa)) + (\mu_g + W_g)\psi(\kappa). \tag{70}$$

Under the null $\boldsymbol{\beta}_{0,g} = \mathbf{0}$, we have $\mu_g = 0$ and

$$\ell_g(\kappa) = \frac{p_g}{2}\phi(\theta(\kappa)) + T_g\psi(\kappa), \qquad T_g := S_g - \frac{p_g}{2}.$$

**Proposition 3.16** (Posterior factorization under the profile specialization)**.** *Assume* (63)*, and suppose further that the priors $\kappa_1, \ldots, \kappa_G$ are independent with $\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa)$. Then, conditional on $(\sigma^2, \tau)$, the joint posterior factorizes across groups:*

$$\Pi(d\kappa_1, \ldots, d\kappa_G \mid \boldsymbol{Y}, \sigma^2, \tau) = \bigotimes_{g=1}^{G} \Pi_g(d\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau). \tag{71}$$

*In particular, every posterior quantity involving only $\kappa_g$ depends only on $\boldsymbol{Y}_g$.*

*Proof.* Under (63), for each group $g$ and each $j \in \mathcal{G}_g$,

$$Y_j \mid \kappa_g, \sigma^2, \tau \sim \mathcal{N}\big(0, \sigma^2(1 + \theta(\kappa_g))\big),$$

and the likelihood contribution of group $g$ depends only on $(\kappa_g, \boldsymbol{Y}_g)$. Since the prior on $(\kappa_1, \ldots, \kappa_G)$ factorizes by assumption, the posterior kernel factorizes as a product of groupwise kernels, which yields (71). □

**Remark 3.17** (Exact but nonconjugate one-dimensional posterior)**.** *Under the profile specialization, each groupwise posterior $\Pi_g(d\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau)$ admits an exact one-dimensional density on $(0, 1)$, proportional to the Beta prior density multiplied by $\exp\{\ell_g(\kappa)\}$. This posterior is exact and one-dimensional, but it does not in general belong to a standard conjugate family.*

**Lemma 3.18** (Local profile geometry)**.** *There exist constants $c_1, c_2, c_3, c_4 > 0$ and $\kappa_0 \in (0,1)$, depending only on $\rho$, such that for all $\kappa \in [0, \kappa_0]$,*

$$c_1\kappa \le \theta(\kappa) \le c_2\kappa, \tag{72}$$

$$c_1\kappa \le \psi(\kappa) \le c_2\kappa, \tag{73}$$

$$-c_4\kappa^2 \le \phi(\theta(\kappa)) \le -c_3\kappa^2. \tag{74}$$

*Moreover, there exists $d_0 > 0$ such that*

$$\phi(\theta(\kappa)) \le -d_0 \qquad \text{for all } \kappa \in [\kappa_0, 1]. \tag{75}$$

*Proof.* Because $\theta(0) = \psi(0) = 0$ and $\theta'(0) = \psi'(0) = 1$, continuity of $\theta(\kappa)/\kappa$ and $\psi(\kappa)/\kappa$ at 0 yields (72) and (73). Since $\phi(x) = -x^2/2 + O(x^3)$ as $x \downarrow 0$, combining this with (72) yields (74). Finally, $\phi(x) < 0$ for every $x > 0$, so continuity on the compact interval $[\kappa_0, 1]$ implies (75). □

**Lemma 3.19** (Null fluctuation)**.** *Under $\boldsymbol{\beta}_{0,g} = \mathbf{0}$, there exists an absolute constant $c_0 > 0$ such that for every $t > 0$,*

$$\mathbb{P}_{\mathbf{0}}\big(|T_g| > t\big) \le 2\exp\left(-c_0 \min\left\{\frac{t^2}{p_g},\, t\right\}\right). \tag{76}$$

*Proof.* Under the null,

$$\frac{Y_j}{\sigma} = \xi_j, \qquad \xi_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

and

$$2T_g = \sum_{j \in \mathcal{G}_g} (\xi_j^2 - 1).$$

Let $X := \xi^2 - 1$ with $\xi \sim \mathcal{N}(0,1)$. For $|t| < 1/2$,

$$\mathbb{E}e^{tX} = e^{-t}(1-2t)^{-1/2},$$

hence

$$\log \mathbb{E}e^{tX} = -t - \frac{1}{2}\log(1-2t) \le \frac{Ct^2}{1-2|t|}$$

for an absolute constant $C > 0$. Summing over $j \in \mathcal{G}_g$ gives

$$\log \mathbb{E}e^{t(2T_g)} \le \frac{Cp_g t^2}{1-2|t|}, \qquad |t| < \frac{1}{2}.$$

Thus $2T_g$ is sub-gamma with variance proxy of order $p_g$ and scale of order one. The standard Bernstein tail bound for centered sub-gamma variables yields (76). □

**Lemma 3.20** (Signal fluctuation)**.** *There exists an absolute constant $c_{\text{sig}} > 0$ such that for every $t > 0$,*

$$\mathbb{P}_{\boldsymbol{\beta}_{0,g}}\big(|W_g| > t\big) \le 2\exp\left(-c_{\text{sig}} \min\left\{\frac{t^2}{p_g + \mu_g},\, t\right\}\right). \tag{77}$$

*Consequently,*

$$W_g = O_{\mathbb{P}_{\boldsymbol{\beta}_{0,g}}}\big(\sqrt{p_g + \mu_g}\big).$$

*In particular,*

$$\frac{\mu_g}{\sqrt{p_g}} \to \infty \quad \Longrightarrow \quad \frac{W_g}{\mu_g} \to 0 \qquad \textit{in } \mathbb{P}_{\boldsymbol{\beta}_{0,g}}\textit{-probability},$$

*and*

$$\mu_g = O(p_g) \quad \Longrightarrow \quad \frac{W_g}{p_g} \to 0 \qquad \textit{in } \mathbb{P}_{\boldsymbol{\beta}_{0,g}}\textit{-probability}.$$

*Proof.* Write

$$\frac{Y_j}{\sigma} = \delta_j + \xi_j, \qquad \delta_j := \frac{\beta_{0,j}}{\sigma}, \qquad \xi_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

Then

$$2W_g = \sum_{j \in \mathcal{G}_g} X_j, \qquad X_j := 2\delta_j \xi_j + (\xi_j^2 - 1).$$

The variables $X_j$ are independent and centered. For $|t| < 1/2$, a direct Gaussian computation gives

$$\mathbb{E}e^{tX_j} = e^{-t}(1-2t)^{-1/2} \exp\left(\frac{2\delta_j^2 t^2}{1-2t}\right),$$

hence

$$\log \mathbb{E}e^{tX_j} = -t - \frac{1}{2}\log(1-2t) + \frac{2\delta_j^2 t^2}{1-2t}.$$

Using

$$-t - \frac{1}{2}\log(1-2t) \le \frac{C_1 t^2}{1-2|t|} \qquad (|t| < 1/2),$$

we obtain

$$\log \mathbb{E}e^{tX_j} \le \frac{C_2(1+\delta_j^2)t^2}{1-2|t|}, \qquad |t| < \frac{1}{2},$$

for an absolute constant $C_2 > 0$. Summing over $j \in \mathcal{G}_g$ yields

$$\log \mathbb{E}e^{t(2W_g)} \le \frac{C_2\big(p_g + \sum_{j \in \mathcal{G}_g} \delta_j^2\big)t^2}{1-2|t|}, \qquad |t| < \frac{1}{2}.$$

Since

$$\sum_{j \in \mathcal{G}_g} \delta_j^2 = \frac{\|\boldsymbol{\beta}_{0,g}\|_2^2}{\sigma^2} = 2\mu_g,$$

we obtain

$$\log \mathbb{E}e^{t(2W_g)} \le \frac{C_3(p_g + \mu_g)t^2}{1-2|t|}, \qquad |t| < \frac{1}{2}.$$

Thus $2W_g$ is sub-gamma with variance proxy of order $p_g + \mu_g$ and scale of order one. Bernstein's inequality for centered sub-gamma variables gives (77). The stated consequences follow immediately. □

**Lemma 3.21** (Beta prior mass near zero)**.** *Suppose*

$$\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \qquad \alpha_\kappa, \beta_\kappa > 0,$$

*and let $\pi_0$ denote the corresponding Beta density. Then:*

(a) *there exist $c_0, s_0 > 0$ such that for all $s \in (0, s_0]$,*

$$\int_0^s \pi_0(\kappa)\, d\kappa \geq c_0 s^{\alpha_\kappa}; \tag{78}$$

(b) *for every $a > 0$, there exist $C_a, s_0 > 0$ such that for all $s \in (0, s_0]$,*

$$\int_0^{as} \pi_0(\kappa)\, d\kappa \leq C_a s^{\alpha_\kappa}; \tag{79}$$

(c) *for every $0 < x_1 < x_2 < \infty$, there exist $C_{x_1,x_2}, s_0 > 0$ such that for all $s \in (0, s_0]$,*

$$\int_{x_1 s}^{x_2 s} \pi_0(\kappa)\, d\kappa \geq C_{x_1,x_2} s^{\alpha_\kappa}. \tag{80}$$

*Proof.* All three statements follow from the Beta density expansion near zero:

$$\pi_0(\kappa) \asymp \kappa^{\alpha_\kappa - 1} \qquad \text{as } \kappa \downarrow 0.$$

Integrating over the corresponding intervals yields the stated bounds. □

## 3.3 Null groups: posterior contraction of the allowance parameter

**Theorem 3.22** (Null-group posterior contraction)**.** *Assume the profile specialization* (63)*, and consider any sequence of null groups $g = g_n$ such that*

$$\boldsymbol{\beta}_{0,g} = \mathbf{0}, \qquad p_g \to \infty.$$

*Let*

$$\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \qquad \alpha_\kappa, \beta_\kappa > 0.$$

*Then:*

(a) *for every deterministic sequence $M_g \to \infty$,*

$$\Pi\left(\kappa_g > \frac{M_g}{\sqrt{p_g}} \;\middle|\; \boldsymbol{Y}_g, \sigma^2, \tau\right) \to 0 \qquad \textit{in } \mathbb{P}_{\mathbf{0}}\textit{-probability}; \tag{81}$$

(b)

$$\mathbb{E}(\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau) = O_{\mathbb{P}_{\mathbf{0}}}(p_g^{-1/2}). \tag{82}$$

*Proof.* Let $\pi_0$ denote the Beta$(\alpha_\kappa, \beta_\kappa)$ prior density and define

$$h_g(\kappa) := \pi_0(\kappa) e^{\ell_g(\kappa)},$$

where under the null

$$\ell_g(\kappa) = \frac{p_g}{2}\phi(\theta(\kappa)) + T_g \psi(\kappa).$$

Fix $\delta \in (0,1)$. By Lemma 3.19, there exists $A_\delta > 0$ such that

$$\mathbb{P}_{\mathbf{0}}\big(|T_g| > A_\delta \sqrt{p_g}\big) \le \delta$$

for all sufficiently large $g$. Define the good event

$$\mathcal{A}_{g,\delta} := \big\{|T_g| \le A_\delta \sqrt{p_g}\big\}.$$

For all sufficiently large $g$, we have $p_g^{-1/2} \le \kappa_0$. On $\mathcal{A}_{g,\delta}$, the denominator admits a lower bound on $[0, p_g^{-1/2}]$. Indeed, by Lemma 3.18,

$$\ell_g(\kappa) \ge -C_1 p_g \kappa^2 - C_2 |T_g| \kappa \ge -C_\delta \qquad \text{for } 0 \le \kappa \le p_g^{-1/2},$$

where $C_\delta < \infty$ depends on $\delta$ but not on $g$. Hence

$$D_g := \int_0^1 h_g(\kappa)\, d\kappa \ge \int_0^{p_g^{-1/2}} h_g(\kappa)\, d\kappa \ge c_\delta p_g^{-\alpha_\kappa/2} \tag{83}$$

by Lemma 3.21(a).

For part (a), let $M_g \to \infty$, and split the posterior numerator over

$$\left[\frac{M_g}{\sqrt{p_g}}, \kappa_0\right] \quad \text{and} \quad [\kappa_0, 1].$$

For $\kappa \in [M_g/\sqrt{p_g}, \kappa_0]$, Lemma 3.18 gives, on $\mathcal{A}_{g,\delta}$,

$$\ell_g(\kappa) \le -c_3 p_g \kappa^2 + c_2 A_\delta \sqrt{p_g}\, \kappa.$$

Choose $g$ sufficiently large that $M_g \ge 2c_2 A_\delta / c_3$. Then

$$-c_3 p_g \kappa^2 + c_2 A_\delta \sqrt{p_g}\, \kappa \le -\frac{c_3}{2} p_g \kappa^2 \qquad \text{for all } \kappa \ge \frac{M_g}{\sqrt{p_g}}.$$

Therefore

$$\begin{aligned} N_{g,1}(M_g) &:= \int_{M_g/\sqrt{p_g}}^{\kappa_0} h_g(\kappa)\, d\kappa \\ &\le C \int_{M_g/\sqrt{p_g}}^{\kappa_0} \kappa^{\alpha_\kappa - 1} e^{-(c_3/2) p_g \kappa^2}\, d\kappa \\ &= C p_g^{-\alpha_\kappa/2} \int_{M_g}^{\kappa_0 \sqrt{p_g}} u^{\alpha_\kappa - 1} e^{-(c_3/2) u^2}\, du. \end{aligned} \tag{84}$$

Since $M_g \to \infty$, the integral on the right converges to zero. Hence

$$\frac{N_{g,1}(M_g)}{D_g} \to 0$$

on $\mathcal{A}_{g,\delta}$.

For $\kappa \in [\kappa_0, 1]$, Lemma 3.18 yields

$$\phi(\theta(\kappa)) \le -d_0.$$

Moreover, the global bound $0 \le \psi(\kappa) \le 1$ gives, on $\mathcal{A}_{g,\delta}$,

$$\ell_g(\kappa) \le -\frac{d_0}{2} p_g + |T_g| \le -\frac{d_0}{2} p_g + A_\delta \sqrt{p_g} \le -c'_\delta p_g$$

for all sufficiently large $g$. Therefore

$$N_{g,2} := \int_{\kappa_0}^{1} h_g(\kappa)\, d\kappa \le C e^{-c'_\delta p_g},$$

so $N_{g,2}/D_g \to 0$ on $\mathcal{A}_{g,\delta}$.

Thus, on $\mathcal{A}_{g,\delta}$,

$$\Pi\left(\kappa_g > \frac{M_g}{\sqrt{p_g}} \,\middle|\, \boldsymbol{Y}_g, \sigma^2, \tau\right) \to 0.$$

Since $\delta$ is arbitrary and $\mathbb{P}_{\mathbf{0}}(\mathcal{A}_{g,\delta}) \ge 1 - \delta$ for all sufficiently large $g$, this proves (81).

For part (b), fix a sufficiently large constant $M > 0$, independent of $g$, and split

$$\mathbb{E}(\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau) = \frac{N_{g,1}^{(0)} + N_{g,2}^{(0)} + N_{g,3}^{(0)}}{D_g},$$

where

$$\begin{aligned} N_{g,1}^{(0)} &:= \int_0^{M/\sqrt{p_g}} \kappa h_g(\kappa)\, d\kappa, \\ N_{g,2}^{(0)} &:= \int_{M/\sqrt{p_g}}^{\kappa_0} \kappa h_g(\kappa)\, d\kappa, \\ N_{g,3}^{(0)} &:= \int_{\kappa_0}^{1} \kappa h_g(\kappa)\, d\kappa. \end{aligned}$$

On $\mathcal{A}_{g,\delta}$,

$$\frac{N_{g,1}^{(0)}}{D_g} \le \frac{M}{\sqrt{p_g}}.$$

Also, using the same local upper bound as above,

$$\begin{aligned} N_{g,2}^{(0)} &\le C \int_{M/\sqrt{p_g}}^{\kappa_0} \kappa^{\alpha_\kappa} e^{-(c_3/2) p_g \kappa^2}\, d\kappa \\ &= C p_g^{-(\alpha_\kappa+1)/2} \int_M^{\kappa_0 \sqrt{p_g}} u^{\alpha_\kappa} e^{-(c_3/2) u^2}\, du \le C'_\delta p_g^{-(\alpha_\kappa+1)/2}. \end{aligned}$$

Dividing by (83) gives

$$\frac{N_{g,2}^{(0)}}{D_g} \le C''_\delta p_g^{-1/2}.$$

Finally,

$$N_{g,3}^{(0)} \le \int_{\kappa_0}^{1} h_g(\kappa)\, d\kappa \le C e^{-c'_\delta p_g},$$

so

$$\frac{N_{g,3}^{(0)}}{D_g} = o(p_g^{-1/2})$$

on $\mathcal{A}_{g,\delta}$. Hence, on $\mathcal{A}_{g,\delta}$,

$$\mathbb{E}(\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau) \le C_\delta p_g^{-1/2}.$$

Since $\mathbb{P}_{\mathbf{0}}(\mathcal{A}_{g,\delta}) \ge 1-\delta$ for all sufficiently large $g$, this proves (82). □

**Corollary 3.23** (Null-group shrinkage concentration)**.** *Under the assumptions of Theorem 3.22,*

$$\max_{j\in\mathcal{G}_g} \mathbb{E}(r_{j,g} \mid \boldsymbol{Y}_g, \sigma^2, \tau) \le \mathbb{E}(\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau) = O_{\mathbb{P}_{\mathbf{0}}}(p_g^{-1/2}), \tag{85}$$

*and hence*

$$\frac{1}{p_g}\sum_{j\in\mathcal{G}_g} \mathbb{E}(r_{j,g} \mid \boldsymbol{Y}_g, \sigma^2, \tau) \to 0 \qquad \text{in } \mathbb{P}_{\mathbf{0}}\text{-probability.} \tag{86}$$

*Proof.* Under the profile specialization,

$$r_{j,g} = \frac{v(\kappa_g)}{\sigma^2 + v(\kappa_g)} = \psi(\kappa_g) \le \kappa_g$$

for every $j \in \mathcal{G}_g$. Taking posterior expectations gives (85). Averaging over $j$ yields (86). □

**Corollary 3.24** (Null-group posterior mean control)**.** *Under the profile specialization and the assumptions of Theorem 3.22,*

$$\|\mathbb{E}(\boldsymbol{\beta}_g \mid \boldsymbol{Y}_g, \sigma^2, \tau)\|_2 \le \mathbb{E}(\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau)\, \|\boldsymbol{Y}_g\|_2, \tag{87}$$

*and therefore*

$$\|\mathbb{E}(\boldsymbol{\beta}_g \mid \boldsymbol{Y}_g, \sigma^2, \tau)\|_2^2 = O_{\mathbb{P}_{\mathbf{0}}}(\sigma^2). \tag{88}$$

*Proof.* The exact bridge inequality (46) remains valid under the profile specialization. Under the null,

$$\|\boldsymbol{Y}_g\|_2^2 = O_{\mathbb{P}_{\mathbf{0}}}(\sigma^2 p_g),$$

while $\mathbb{E}(\kappa_g \mid \boldsymbol{Y}_g, \sigma^2, \tau)^2 = O_{\mathbb{P}_{\mathbf{0}}}(p_g^{-1})$ by Theorem 3.22. Multiplying the two factors yields (88). □

## 3.4 Signal groups: escape, adaptive localization, and retention

**Proposition 3.25** (Local signal-side profile bounds)**.** *There exist constants $A_1, A_2, A_3, B_1, B_2, B_3 > 0$, depending only on $\rho$, such that for all $\kappa \in [0, \kappa_0]$,*

$$-A_1 p_g \kappa^2 + A_2 \mu_g \kappa - A_3 |W_g| \kappa \le \ell_g(\kappa) \le -B_1 p_g \kappa^2 + B_2 \mu_g \kappa + B_3 |W_g| \kappa. \tag{89}$$

*Proof.* Combine (70) with the local geometry bounds (73) and (74). □

**Lemma 3.26** (Rescaled local profile)**.** *Define*

$$s_g := \frac{\mu_g}{p_g}, \qquad r_g := \frac{\mu_g^2}{p_g}.$$

*Then there exist constants $a_1, a_2, a_3, b_1, b_2, b_3 > 0$, depending only on $\rho$, such that whenever $x s_g \le \kappa_0$,*

$$\ell_g(x s_g) \ge r_g\big(-a_1 x^2 + a_2 x\big) - a_3 x \frac{|W_g|}{\mu_g} r_g, \tag{90}$$

$$\ell_g(x s_g) \le r_g\big(-b_1 x^2 + b_2 x\big) + b_3 x \frac{|W_g|}{\mu_g} r_g. \tag{91}$$

*Proof.* Substitute $\kappa = x\mu_g/p_g$ into (89). Since

$$p_g \kappa^2 = r_g x^2, \qquad \mu_g \kappa = r_g x,$$

the claim follows. □

**Theorem 3.27** (Signal-side escape from the null scale)**.** *Assume the profile specialization and suppose, along a sequence of signal groups $g = g_n$,*

$$\frac{\mu_g}{\sqrt{p_g}} \to \infty, \qquad \frac{\mu_g}{p_g} \to 0. \tag{92}$$

*Then, for every fixed $M > 0$,*

$$\Pi\left(\kappa_g \le \frac{M}{\sqrt{p_g}} \;\middle|\; \boldsymbol{Y}_g, \sigma^2, \tau\right) \to 0 \qquad \text{in } \mathbb{P}_{\boldsymbol{\beta}_{0,g}}\text{-probability.} \tag{93}$$

*Proof.* Let

$$s_g := \frac{\mu_g}{p_g}, \qquad r_g := \frac{\mu_g^2}{p_g}.$$

Then $s_g \to 0$ and $r_g \to \infty$.

Fix $\eta \in (0, 1)$ and define

$$E_{g,\eta} := \{|W_g| \le \eta \mu_g\}.$$

By Lemma 3.20,

$$\mathbb{P}_{\boldsymbol{\beta}_{0,g}}(E_{g,\eta}) \to 1.$$

For all sufficiently large $g$, we also have $M/\sqrt{p_g} \le \kappa_0$. On the small interval $[0, M/\sqrt{p_g}]$, Proposition 3.25 yields, on $E_{g,\eta}$,

$$\ell_g(\kappa) \le -B_1 p_g \kappa^2 + B_2 \mu_g \kappa + B_3 |W_g| \kappa \le C_1 \frac{\mu_g}{\sqrt{p_g}}.$$

Therefore

$$N_g^{\text{small}} := \int_0^{M/\sqrt{p_g}} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa \le C_2 p_g^{-\alpha_\kappa/2} \exp\left( C_1 \frac{\mu_g}{\sqrt{p_g}} \right) \tag{94}$$

by Lemma 3.21(b).

Now choose $A_\star > 0$ small enough and consider the interval

$$I_g := [A_\star s_g,\, 2A_\star s_g].$$

For all sufficiently large $g$, this interval lies inside $(0, \kappa_0]$. By Lemma 3.26, for $\eta > 0$ sufficiently small,

$$\inf_{\kappa \in I_g} \ell_g(\kappa) \ge c_\star r_g$$

on $E_{g,\eta}$, for some $c_\star > 0$. Hence

$$D_g^{\text{mid}} := \int_{I_g} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa \ge C_3 s_g^{\alpha_\kappa} e^{c_\star r_g} \tag{95}$$

by Lemma 3.21(c).

Combining (94) and (95), we obtain on $E_{g,\eta}$,

$$\frac{N_g^{\text{small}}}{D_g^{\text{mid}}} \le C_4 \frac{p_g^{-\alpha_\kappa/2}}{s_g^{\alpha_\kappa}} \exp\left( C_1 \frac{\mu_g}{\sqrt{p_g}} - c_\star r_g \right).$$

Since

$$\frac{p_g^{-\alpha_\kappa/2}}{s_g^{\alpha_\kappa}} = \frac{p_g^{\alpha_\kappa/2}}{\mu_g^{\alpha_\kappa}},$$

and

$$r_g = \frac{\mu_g^2}{p_g} \to \infty, \qquad \frac{\mu_g}{\sqrt{p_g}} = o(r_g),$$

the right-hand side tends to zero. Because the full denominator dominates $D_g^{\text{mid}}$, the posterior probability of $[0, M/\sqrt{p_g}]$ tends to zero on $E_{g,\eta}$, hence in $\mathbb{P}_{\boldsymbol{\beta}_{0,g}}$-probability. □

**Lemma 3.28** (Two-sided quadratic window separation)**.** *There exists $\eta_0 > 0$ such that for every $\eta \in (0, \eta_0)$, there exist constants*

$$0 < x_- < x_1 < x_2 < x_+ < \infty, \qquad m, c, d > 0,$$

*depending only on $\rho$ and $\eta$, such that*

$$\inf_{x \in [x_1, x_2]} \big( -a_1 x^2 + (a_2 - a_3 \eta) x \big) \ge m, \tag{96}$$

$$\sup_{x \in [0, x_-]} \big( -b_1 x^2 + (b_2 + b_3 \eta) x \big) \le m - c, \tag{97}$$

$$-b_1 x^2 + (b_2 + b_3 \eta) x \le -d x^2 \qquad \textit{for all } x \ge x_+. \tag{98}$$

*Proof.* Choose $\eta_0 > 0$ so small that $a_2 - a_3\eta > 0$ for all $\eta \in (0, \eta_0)$. Then the quadratic

$$q_L(x) := -a_1 x^2 + (a_2 - a_3\eta)x$$

is strictly concave, positive on some compact interval in $(0, \infty)$, and attains a positive maximum at

$$x_L^\star = \frac{a_2 - a_3\eta}{2a_1} > 0.$$

Choose a compact interval $[x_1, x_2] \subset (0, \infty)$ containing $x_L^\star$ such that

$$\inf_{x \in [x_1, x_2]} q_L(x) \geq m > 0.$$

Next, since

$$q_U(x) := -b_1 x^2 + (b_2 + b_3\eta)x$$

is continuous and $q_U(0) = 0$, choose $x_- \in (0, x_1)$ so small that

$$\sup_{x \in [0, x_-]} q_U(x) \leq m - c$$

for some $c \in (0, m)$. Finally, because the leading coefficient of $q_U$ is $-b_1 < 0$, we have $q_U(x) \to -\infty$ as $x \to \infty$. Hence there exist $x_+ > x_2$ and $d > 0$ such that

$$q_U(x) \leq -dx^2 \qquad \text{for all } x \geq x_+.$$

□

**Lemma 3.29** (Gaussian tail integral bound)**.** *Fix $\alpha > 0$, $d > 0$, and $A_0 > 0$. Then there exist constants $C, c > 0$ such that for every $A \geq A_0$,*

$$\int_A^\infty t^{\alpha-1} e^{-dt^2}\, dt \leq C e^{-cA^2}. \tag{99}$$

*Proof.* Polynomial growth is dominated by Gaussian decay. Hence there exists $A_0' \geq A_0$ such that

$$t^{\alpha-1} \leq e^{dt^2/2} \qquad \text{for all } t \geq A_0'.$$

Therefore, for $A \geq A_0'$,

$$\int_A^\infty t^{\alpha-1} e^{-dt^2}\, dt \leq \int_A^\infty e^{-dt^2/2}\, dt \leq C_1 e^{-dA^2/4}.$$

Enlarging the constant if necessary to cover the compact range $A \in [A_0, A_0']$ proves the claim. □

**Theorem 3.30** (Adaptive localization on the scale $\mu_g/p_g$)**.** *Assume the profile specialization and the intermediate regime* (92)*. Then there exist fixed constants*

$$0 < x_- < x_+ < \infty,$$

*depending only on $\rho$ and $(\alpha_\kappa, \beta_\kappa)$, such that*

$$\Pi\left( x_- \frac{\mu_g}{p_g} \leq \kappa_g \leq x_+ \frac{\mu_g}{p_g} \;\middle|\; \boldsymbol{Y}_g, \sigma^2, \tau \right) \to 1 \qquad \text{in } \mathbb{P}_{\boldsymbol{\beta}_{0,g}}\text{-probability.} \tag{100}$$

*Proof.* Let

$$s_g := \frac{\mu_g}{p_g}, \qquad r_g := \frac{\mu_g^2}{p_g}.$$

By (92), $s_g \to 0$ and $r_g \to \infty$.

Choose $\eta \in (0, \eta_0)$, where $\eta_0$ is the constant from Lemma 3.28, and define

$$E_{g,\eta} := \{|W_g| \le \eta \mu_g\}.$$

By Lemma 3.20,

$$\mathbb{P}_{\boldsymbol{\beta}_{0,g}}(E_{g,\eta}) \to 1.$$

Let $x_-, x_1, x_2, x_+, m, c, d$ be the constants from Lemma 3.28. Since $s_g \to 0$, we have $x_+ s_g \le \kappa_0$ for all sufficiently large $g$.

Define the middle interval

$$J_g := [x_1 s_g,\, x_2 s_g].$$

On $E_{g,\eta}$, Lemma 3.26 together with (96) yields

$$\inf_{\kappa \in J_g} \ell_g(\kappa) \ge m r_g.$$

Hence

$$D_g^{\text{mid}} := \int_{J_g} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa \ge C_J s_g^{\alpha_\kappa} e^{m r_g} \tag{101}$$

by Lemma 3.21(c).

Next consider the lower tail

$$N_g^{\text{low}} := \int_0^{x_- s_g} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa.$$

On $E_{g,\eta}$, Lemma 3.26 and (97) imply

$$\sup_{0 \le \kappa \le x_- s_g} \ell_g(\kappa) \le (m - c) r_g.$$

Hence

$$N_g^{\text{low}} \le C_- s_g^{\alpha_\kappa} e^{(m-c) r_g}$$

by Lemma 3.21(b), and therefore

$$\frac{N_g^{\text{low}}}{D_g^{\text{mid}}} \le C e^{-c r_g} \to 0$$

on $E_{g,\eta}$.

Now consider the local upper tail

$$N_g^{\text{high,loc}} := \int_{x_+ s_g}^{\kappa_0} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa.$$

On $E_{g,\eta}$, Lemma 3.26 and (98) imply

$$\ell_g(\kappa) \le -dr_g \left(\frac{\kappa}{s_g}\right)^2 \qquad \text{for } x_+ s_g \le \kappa \le \kappa_0.$$

Hence, using $\pi_0(\kappa) \le C\kappa^{\alpha_\kappa - 1}$ on $(0, \kappa_0]$,

$$\begin{aligned} N_g^{\text{high,loc}} &\le C \int_{x_+ s_g}^{\kappa_0} \kappa^{\alpha_\kappa - 1} e^{-dr_g(\kappa/s_g)^2}\, d\kappa \\ &= C s_g^{\alpha_\kappa} \int_{x_+}^{\kappa_0/s_g} x^{\alpha_\kappa - 1} e^{-dr_g x^2}\, dx. \end{aligned}$$

Set $t = x\sqrt{r_g}$. Then

$$N_g^{\text{high,loc}} \le C s_g^{\alpha_\kappa} r_g^{-\alpha_\kappa/2} \int_{x_+\sqrt{r_g}}^{\infty} t^{\alpha_\kappa - 1} e^{-dt^2}\, dt.$$

By Lemma 3.29, the last integral is bounded by $C' e^{-c' r_g}$. Therefore

$$\frac{N_g^{\text{high,loc}}}{D_g^{\text{mid}}} \le C'' r_g^{-\alpha_\kappa/2} e^{-(m+c')r_g} \to 0$$

on $E_{g,\eta}$.

Finally, consider the far tail

$$N_g^{\text{far}} := \int_{\kappa_0}^{1} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa.$$

Since $\mu_g/p_g \to 0$, we also have $W_g/p_g \to 0$ by Lemma 3.20. Hence, with probability tending to one,

$$\sup_{\kappa \in [\kappa_0, 1]} \ell_g(\kappa) \le -c_0 p_g$$

for some $c_0 > 0$, by (75). Consequently

$$N_g^{\text{far}} \le C e^{-c_0 p_g},$$

and so

$$\frac{N_g^{\text{far}}}{D_g^{\text{mid}}} \to 0$$

in $\mathbb{P}_{\boldsymbol{\beta}_{0,g}}$-probability.

Combining the three tail bounds yields

$$\Pi\big(\kappa_g \notin [x_- s_g,\, x_+ s_g] \,\big|\, \boldsymbol{Y}_g, \sigma^2, \tau\big) \to 0$$

in $\mathbb{P}_{\boldsymbol{\beta}_{0,g}}$-probability, which proves (100). □

To describe the strong-signal regime, define

$$F_\xi(\kappa) := \frac{1}{2}\phi(\theta(\kappa)) + \xi\psi(\kappa), \qquad \xi \geq 0. \tag{102}$$

Then

$$\ell_g(\kappa) = p_g F_{\mu_g/p_g}(\kappa) + W_g\psi(\kappa).$$

**Lemma 3.31** (Profile phase diagram)**.** *For every $\xi \geq 0$ and $\kappa \in (0,1)$,*

$$F'_\xi(\kappa) = \psi'(\kappa)\left(\xi - \frac{\theta(\kappa)}{2}\right). \tag{103}$$

*Moreover, $\psi'(\kappa) > 0$ and $\theta(\kappa)$ is strictly increasing on $(0,1)$, with*

$$\theta(0) = 0, \qquad \theta(1) = \rho^2.$$

*Consequently:*

(a) *if $0 \leq \xi < \rho^2/2$, then $F_\xi$ has a unique interior maximizer $\kappa_\star(\xi) \in (0,1)$, characterized by*

$$\theta(\kappa_\star(\xi)) = 2\xi;$$

(b) *if $\xi \geq \rho^2/2$, then $F_\xi$ is increasing on $[0,1]$, and its maximum over $[0,1]$ is attained at $\kappa = 1$.*

*Proof.* Since

$$\psi(\kappa) = \frac{\theta(\kappa)}{1+\theta(\kappa)},$$

we have

$$\psi'(\kappa) = \frac{\theta'(\kappa)}{(1+\theta(\kappa))^2},$$

while $\phi'(x) = -x/(1+x)^2$. Therefore

$$F'_\xi(\kappa) = \frac{1}{2}\phi'(\theta(\kappa))\theta'(\kappa) + \xi\psi'(\kappa) = \psi'(\kappa)\left(\xi - \frac{\theta(\kappa)}{2}\right).$$

The remaining statements follow because $\psi'(\kappa) > 0$ and $\theta(\kappa)$ is strictly increasing from 0 to $\rho^2$. □

**Theorem 3.32** (Strong-signal posterior retention)**.** *Fix $u_0 \in (0,1)$, and define*

$$\xi_{\mathrm{crit}}(u_0,\rho) := \frac{\theta(u_0)}{2} = \frac{u_0\rho^2}{2\big(u_0 + (1-u_0)\rho^2\big)}. \tag{104}$$

*Assume the profile specialization and suppose, along a sequence of signal groups $g = g_n$,*

$$p_g \to \infty, \qquad \frac{\mu_g}{p_g} \to \xi_\infty, \qquad \xi_\infty > \xi_{\mathrm{crit}}(u_0,\rho).$$

*Then*

$$\Pi(\kappa_g > u_0 \mid \boldsymbol{Y}_g, \sigma^2, \tau) \to 1 \qquad \text{in } \mathbb{P}_{\boldsymbol{\beta}_{0,g}}\text{-probability.} \tag{105}$$

*Proof.* We first identify the deterministic profile gap. If $\xi_\infty \geq \rho^2/2$, then by Lemma 3.31, $F_{\xi_\infty}$ is strictly increasing on $[0,1]$, so

$$\sup_{\kappa\in[0,u_0]} F_{\xi_\infty}(\kappa) = F_{\xi_\infty}(u_0).$$

If instead $0 \leq \xi_\infty < \rho^2/2$, then $F_{\xi_\infty}$ has a unique interior maximizer $\kappa_\star(\xi_\infty)$, characterized by

$$\theta(\kappa_\star(\xi_\infty)) = 2\xi_\infty.$$

Because $\xi_\infty > \theta(u_0)/2$ and $\theta$ is strictly increasing, we have $\kappa_\star(\xi_\infty) > u_0$, so $F_{\xi_\infty}$ is strictly increasing on $[0,u_0]$, and again

$$\sup_{\kappa\in[0,u_0]} F_{\xi_\infty}(\kappa) = F_{\xi_\infty}(u_0).$$

If $0 \leq \xi_\infty < \rho^2/2$, then the unique maximizer $\kappa_\star(\xi_\infty)$ lies strictly to the right of $u_0$. By continuity of $F_{\xi_\infty}$, there exists a compact interval $J = [u_1, u_2] \subset (u_0, 1)$ containing $\kappa_\star(\xi_\infty)$ and a constant $c_\star > 0$ such that

$$\inf_{\kappa\in J} F_{\xi_\infty}(\kappa) - \sup_{\kappa\in[0,u_0]} F_{\xi_\infty}(\kappa) \geq c_\star.$$

If $\xi_\infty \geq \rho^2/2$, then $F_{\xi_\infty}$ is strictly increasing on $[0,1]$. Hence, again by continuity, there exists a compact interval $J = [u_1, u_2] \subset (u_0, 1)$, chosen sufficiently close to 1, and a constant $c_\star > 0$ such that

$$\inf_{\kappa\in J} F_{\xi_\infty}(\kappa) - \sup_{\kappa\in[0,u_0]} F_{\xi_\infty}(\kappa) \geq c_\star.$$

Because

$$F_{\mu_g/p_g}(\kappa) - F_{\xi_\infty}(\kappa) = \left(\frac{\mu_g}{p_g} - \xi_\infty\right)\psi(\kappa),$$

and $\psi$ is bounded on $[0,1]$, we have

$$\sup_{\kappa\in[0,1]} \left|F_{\mu_g/p_g}(\kappa) - F_{\xi_\infty}(\kappa)\right| \to 0.$$

Hence, for all sufficiently large $g$,

$$\inf_{\kappa\in J} F_{\mu_g/p_g}(\kappa) - \sup_{\kappa\in[0,u_0]} F_{\mu_g/p_g}(\kappa) \geq \frac{c_\star}{2}.$$

Since $\mu_g/p_g \to \xi_\infty < \infty$, we have $\mu_g = O(p_g)$, and Lemma 3.20 gives

$$\frac{W_g}{p_g} \to 0 \qquad \text{in } \mathbb{P}_{\boldsymbol{\beta}_{0,g}}\text{-probability.}$$

Now define

$$E_{g,\varepsilon} := \{|W_g| \leq \varepsilon p_g\}$$

with $0 < \varepsilon < c_\star/8$. Then

$$\mathbb{P}_{\boldsymbol{\beta}_{0,g}}(E_{g,\varepsilon}) \to 1.$$

On $E_{g,\varepsilon}$, since $0 \le \psi(\kappa) \le 1$,

$$\sup_{\kappa\in[0,u_0]} \ell_g(\kappa) \le p_g \sup_{\kappa\in[0,u_0]} F_{\mu_g/p_g}(\kappa) + \varepsilon p_g,$$

while

$$\inf_{\kappa\in J} \ell_g(\kappa) \ge p_g \inf_{\kappa\in J} F_{\mu_g/p_g}(\kappa) - \varepsilon p_g.$$

Hence

$$\inf_{\kappa\in J} \ell_g(\kappa) - \sup_{\kappa\in[0,u_0]} \ell_g(\kappa) \ge \frac{c_\star}{4} p_g$$

for all sufficiently large $g$.

Therefore, if

$$N_g := \int_0^{u_0} \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa, \qquad D_g := \int_J \pi_0(\kappa) e^{\ell_g(\kappa)}\, d\kappa,$$

then on $E_{g,\varepsilon}$,

$$\frac{N_g}{D_g} \le C e^{-(c_\star/4)p_g} \to 0,$$

where $C < \infty$ depends only on the Beta prior and $J$. Since the full posterior denominator dominates $D_g$, it follows that

$$\Pi(\kappa_g \le u_0 \mid \boldsymbol{Y}_g, \sigma^2, \tau) \to 0$$

in $\mathbb{P}_{\boldsymbol{\beta}_{0,g}}$-probability. □

**Corollary 3.33** (Phase-diagram interpretation of strong-signal retention)**.** *Let $F_\xi$ be defined by* (102)*.*

*If $0 \le \xi < \rho^2/2$, then $F_\xi$ has the unique interior maximizer*

$$\kappa_\star(\xi) = \frac{2\xi\,\rho^2}{\rho^2 - 2\xi(1-\rho^2)}. \tag{106}$$

*Moreover, for every fixed $u_0 \in (0,1)$,*

$$\xi > \xi_{\text{crit}}(u_0, \rho) \qquad \Longleftrightarrow \qquad \kappa_\star(\xi) > u_0. \tag{107}$$

*If $\xi \ge \rho^2/2$, then $F_\xi$ is increasing on $[0,1]$, so its maximizer is attained at the boundary $\kappa = 1$.*

*Consequently, Theorem 3.32 may be interpreted as a posterior lift of the deterministic phase diagram of $F_\xi$: posterior mass concentrates to the right of $u_0$ precisely when the deterministic profile maximizer lies to the right of $u_0$.*

*Proof.* By Lemma 3.31, if $0 \le \xi < \rho^2/2$, then the unique interior maximizer $\kappa_\star(\xi)$ is characterized by

$$\theta(\kappa_\star(\xi)) = 2\xi.$$

Since

$$\theta(\kappa) = \frac{\kappa\rho^2}{\kappa + (1-\kappa)\rho^2},$$

solving $\theta(\kappa) = 2\xi$ yields

$$\kappa_\star(\xi) = \frac{2\xi\,\rho^2}{\rho^2 - 2\xi(1-\rho^2)}.$$

Because $\theta$ is strictly increasing,

$$\xi > \xi_{\text{crit}}(u_0, \rho) \iff 2\xi > \theta(u_0) \iff \kappa_\star(\xi) > u_0.$$

If $\xi \geq \rho^2/2$, the boundary-maximizer statement follows directly from Lemma 3.31. The final interpretation is therefore immediate from Theorem 3.32. □

## 3.5 Simultaneous groupwise separation and blockwise orthonormal regression

**Theorem 3.34** (Simultaneous separation under fixed $G$)**.** *Assume the profile specialization and suppose the number of groups $G$ is fixed across the underlying sequence. Let $S \subset \{1, \ldots, G\}$ denote the set of signal groups and $S^c$ the set of null groups. Assume throughout that both $S$ and $S^c$ are nonempty.*

*Assume that every null group $g \in S^c$ satisfies the conditions of Theorem 3.22, and every signal group $g \in S$ satisfies the strong-signal retention conditions of Theorem 3.32 for a common threshold $u_0 \in (0,1)$. Then*

$$\max_{g \in S^c} \mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2, \tau) = O_{\mathbb{P}_{\boldsymbol{\beta}_0}}\left(p_{\min,S^c}^{-1/2}\right), \tag{108}$$

*where*

$$p_{\min,S^c} := \min_{g \in S^c} p_g,$$

*and*

$$\min_{g \in S} \Pi(\kappa_g > u_0 \mid \boldsymbol{Y}, \sigma^2, \tau) \to 1 \qquad \textit{in } \mathbb{P}_{\boldsymbol{\beta}_0}\textit{-probability.} \tag{109}$$

*Proof.* By Proposition 3.16, under the profile specialization with fixed $\tau$ and independent Beta priors on the $\kappa_g$'s, every posterior quantity involving only $\kappa_g$ depends only on $\boldsymbol{Y}_g$. For each null group $g \in S^c$, Theorem 3.22 gives

$$\mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2, \tau) = O_{\mathbb{P}}(p_g^{-1/2}).$$

Hence for every $M > 0$,

$$\begin{aligned}\mathbb{P}\Bigg(p_{\min,S^c}^{1/2} \max_{g \in S^c} \mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2, \tau) > M\Bigg) \\ \leq \sum_{g \in S^c} \mathbb{P}\big(p_g^{1/2}\mathbb{E}(\kappa_g \mid \boldsymbol{Y}, \sigma^2, \tau) > M\big) \to 0,\end{aligned}$$

because $G$ is fixed.

Similarly, for each $g \in S$,

$$\Pi(\kappa_g > u_0 \mid \boldsymbol{Y}, \sigma^2, \tau) \to 1$$

by Theorem 3.32. Therefore for every $\varepsilon > 0$,

$$\mathbb{P}\left(\min_{g \in S} \Pi(\kappa_g > u_0 \mid \boldsymbol{Y}, \sigma^2, \tau) \leq 1 - \varepsilon\right) \leq \sum_{g \in S} \mathbb{P}\big(\Pi(\kappa_g > u_0 \mid \boldsymbol{Y}, \sigma^2, \tau) \leq 1 - \varepsilon\big) \to 0.$$

□

**Proposition 3.35** (Blockwise orthonormal grouped regression lift)**.** *Consider the grouped regression model*

$$\boldsymbol{y} = X\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 I_n),$$

*and suppose that the design blocks are mutually orthogonal and individually orthonormal:*

$$X_g^\top X_h = 0 \quad (g \neq h), \qquad X_g^\top X_g = I_{p_g}, \qquad g = 1, \ldots, G.$$

*Define the block statistic*

$$\boldsymbol{W}_g := X_g^\top \boldsymbol{y}.$$

*Then*

$$\boldsymbol{W}_g \sim \mathcal{N}(\boldsymbol{\beta}_{0,g}, \sigma^2 I_{p_g}), \qquad g = 1, \ldots, G,$$

*with independence across groups. Consequently, under the profile specialization* (63) *together with independent Beta priors on the $\kappa_g$'s, the grouped normal-means posterior theory established above applies blockwise to $\boldsymbol{W}_g$. In particular:*

(a) *if $\boldsymbol{\beta}_{0,g} = \boldsymbol{0}$, then*

$$\mathbb{E}(\kappa_g \mid \boldsymbol{y}, \sigma^2, \tau) = O_{\mathbb{P}_{\boldsymbol{\beta}_0}}(p_g^{-1/2});$$

(b) *if $\boldsymbol{\beta}_{0,g}$ satisfies the corresponding intermediate- or strong-signal assumptions, then the conclusions of Theorems 3.30 and 3.32 hold with $\boldsymbol{Y}_g$ replaced by $\boldsymbol{W}_g$.*

*Proof.* Because the blocks are mutually orthogonal and individually orthonormal,

$$\boldsymbol{W}_g = X_g^\top (X_g \boldsymbol{\beta}_{0,g} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta}_{0,g} + X_g^\top \boldsymbol{\varepsilon}.$$

Moreover,

$$X_g^\top \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 I_{p_g}),$$

and the block noises are independent because $X_g^\top X_h = 0$ for $g \neq h$. Hence $\boldsymbol{W}_g$ is exactly a grouped normal-means observation for group $g$. Under the profile specialization with fixed $\tau$, the posterior factorization from Proposition 3.16 applies blockwise, so the preceding posterior theory transfers directly to $\boldsymbol{W}_g$. □

**Remark 3.36** (Scope of the posterior theory)**.** *The exact bridge theorem, posterior propriety, and the conditional posterior geometry in Subsection 3.1 hold under the full grouped regularized horseshoe hierarchy in the grouped normal-means representation. By contrast, the sharp posterior contraction, localization, retention, and simultaneous separation results established in Subsections 3.3– 3.5 are profile-specialization results, and they rely on the one-dimensional specialization $\lambda_j \equiv 1$, $a_g \equiv 1$, with $\tau$ treated as fixed. Under this specialization, the posterior of each $\kappa_g$ is exact and one-dimensional but generally nonconjugate, and the posterior factorizes across groups, which is the mechanism used in the simultaneous separation and blockwise regression lift. These sharp results should therefore be read as posterior theory for the allowance parameter in a canonical one-dimensional specialization of the full grouped hierarchy, rather than as a full posterior contraction theory for the complete model with random local, group-amplitude, and global-scale layers.*

# A Computational representation and conditional structure

## A.1 Primitive hierarchy and computational specialization

The purpose of this appendix is to make the conditional structure of the GR-RHS hierarchy explicit under a concrete full Bayesian specialization. These derivations are exact and remain fully faithful to the primitive model defined in Section 1; the auxiliary variables and transformed quantities introduced below are used only for computation.

Throughout the appendix, $\mathrm{IG}(a, b)$ denotes the inverse-gamma distribution with density

$$f(x \mid a, b) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp(-b/x), \qquad x > 0.$$

We also use the shape–rate convention for gamma distributions.

Throughout this appendix, we specialize the primitive hierarchy to

$$\kappa_g \sim \mathrm{Beta}(\alpha_\kappa, \beta_\kappa), \qquad \sigma^2 \sim \mathrm{IG}(a_\sigma, b_\sigma), \qquad \tau \mid \sigma \sim \mathrm{C}^+\big(0, \tilde{\tau}_0\, \sigma\big).$$

For computational convenience we introduce the deterministic one-to-one transforms

$$u_j := \lambda_j^2, \qquad b_g := a_g^2, \qquad \eta := \tau^2, \qquad q_g := c_g^2 = \frac{\sigma^2 \kappa_g}{1 - \kappa_g}.$$

These are not new primitive parameters. In particular, $q_g$ is only the derived slab-variance variable induced by $\kappa_g$ and $\sigma^2$.

Let $g(j)$ denote the unique group index such that $j \in \mathcal{G}_{g(j)}$. Then the prior variance of $\beta_j$ may be written as

$$v_j = \frac{q_{g(j)}\, \eta\, u_j\, b_{g(j)}}{q_{g(j)} + \eta\, u_j\, b_{g(j)}}. \tag{110}$$

Equivalently,

$$\frac{1}{v_j} = \frac{1}{q_{g(j)}} + \frac{1}{\eta\, u_j\, b_{g(j)}}. \tag{111}$$

Under the transformation $q_g = \sigma^2\kappa_g/(1-\kappa_g)$, the default Beta allowance prior induces the conditional density

$$\pi(q_g \mid \sigma^2) = \frac{(\sigma^2)^{\beta_\kappa}}{B(\alpha_\kappa, \beta_\kappa)} q_g^{\alpha_\kappa - 1}(\sigma^2 + q_g)^{-(\alpha_\kappa+\beta_\kappa)}, \qquad q_g > 0. \tag{112}$$

This makes explicit that, in the $q_g$-parameterization, the induced prior on $q_g$ depends on $\sigma^2$. This is not a model modification; it is the direct consequence of the primitive relation $q_g = \sigma^2\kappa_g/(1-\kappa_g)$.

**Remark A.1.** *This appendix records an exact conditional anatomy of the primitive GR-RHS model under a concrete computational specialization. Some components admit named full conditional distributions, while several others admit only exact one-dimensional conditional kernels. The appendix should therefore be read as a computational decomposition of the primitive model, not as a claim that the full hierarchy is globally Gibbs-closed in elementary families.*

## A.2 Auxiliary-variable representations

To expose the conditional structure of the half-Cauchy layers, we use the standard inverse-gamma mixture representation.

**Local half-Cauchy layer.** For each $j$,

$$u_j \mid \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \qquad \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, 1\right). \tag{113}$$

Marginally, this implies $\lambda_j = \sqrt{u_j} \sim \mathrm{C}^+(0,1)$.

**Global half-Cauchy layer.** For the global scale,

$$\eta \mid \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \qquad \xi \mid \sigma^2 \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\tilde{\tau}_0^2\sigma^2}\right). \tag{114}$$

Marginally, this implies $\tau = \sqrt{\eta} \mid \sigma \sim \mathrm{C}^+\big(0, \tilde{\tau}_0\,\sigma\big)$.

**Group-amplitude layer.** If $a_g \sim \mathrm{HN}(0, s_{a,g}^2)$, then $b_g = a_g^2$ has density

$$\pi(b_g) \propto b_g^{-1/2} \exp\left(-\frac{b_g}{2s_{a,g}^2}\right), \qquad b_g > 0. \tag{115}$$

that is,

$$b_g \sim \mathrm{Gamma}\left(\frac{1}{2}, \frac{1}{2s_{a,g}^2}\right)$$

under the shape–rate convention.

## A.3 Gaussian coefficient block

Conditional on the latent scales, the model remains Gaussian in $\boldsymbol{\beta}$. Let

$$D_v := \mathrm{diag}(v_1, \ldots, v_p),$$

where $v_j$ is given by (110). Then

$$\boldsymbol{y} \mid \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(X\boldsymbol{\beta}, \sigma^2 I_n), \qquad \boldsymbol{\beta} \mid \text{scales}, \sigma^2 \sim \mathcal{N}(\mathbf{0}, D_v).$$

Hence

$$\boldsymbol{\beta} \mid \boldsymbol{y}, \sigma^2, \{u_j\}, \{b_g\}, \{q_g\}, \eta \sim \mathcal{N}(m_\beta, \Sigma_\beta), \tag{116}$$

with

$$\Sigma_\beta = \left(\frac{1}{\sigma^2} X^\top X + D_v^{-1}\right)^{-1}, \qquad m_\beta = \Sigma_\beta \frac{1}{\sigma^2} X^\top \boldsymbol{y}. \tag{117}$$

## A.4 Full conditional distributions and kernels

In this subsection, we distinguish between *exact full conditional distributions* and *exact full conditional kernels*. The former are available in standard named form; the latter are exact up to a proportionality constant but are not, in general, of a standard elementary family. In particular, only the Gaussian coefficient block and a subset of auxiliary variables admit named closed-form conditionals.

### A.4.1 Exact full conditional distributions

**Coefficient block.** The coefficient block is Gaussian as in (116)–(117).

**Local auxiliary variables $\nu_j$.** From (113),

$$\nu_j \mid u_j \sim \mathrm{IG}\left(1,\, 1 + \frac{1}{u_j}\right), \qquad j = 1, \ldots, p. \tag{118}$$

**Global auxiliary variable $\xi$.** From (114),

$$\xi \mid \eta, \sigma^2 \sim \mathrm{IG}\left(1,\, \frac{1}{\eta} + \frac{1}{\tilde{\tau}_0^2 \sigma^2}\right). \tag{119}$$

### A.4.2 Exact full conditional kernels

For the kernels below, terms not involving the variable of interest are suppressed.

**Local variance parameter $u_j = \lambda_j^2$.** Fix $j \in \mathcal{G}_g$, and let $g = g(j)$. Using (110)–(111) and the prior (113), we obtain

$$\pi(u_j \mid \text{rest}) \propto u_j^{-2} \big(q_g + \eta\, b_g\, u_j\big)^{1/2} \exp\left\{-\frac{1}{u_j}\left(\frac{\beta_j^2}{2\eta\, b_g} + \frac{1}{\nu_j}\right)\right\}, \qquad u_j > 0. \tag{120}$$

**Group-amplitude variance parameter $b_g = a_g^2$.** Let $\mathcal{G}_g$ denote group $g$. From (115) and the Gaussian prior factors for $\{\beta_j : j \in \mathcal{G}_g\}$,

$$\pi(b_g \mid \text{rest}) \propto b_g^{-(p_g+1)/2} \prod_{j\in\mathcal{G}_g} \big(q_g+\eta\, u_j\, b_g\big)^{1/2} \exp\left\{-\frac{b_g}{2s_{a,g}^2} - \sum_{j\in\mathcal{G}_g} \frac{\beta_j^2}{2\eta\, u_j\, b_g}\right\}, \qquad b_g > 0. \tag{121}$$

**Global variance parameter $\eta = \tau^2$.** Combining the Gaussian prior factors for $\boldsymbol{\beta}$ with the first-stage global auxiliary prior in (114),

$$\pi(\eta \mid \text{rest}) \propto \eta^{-(p+3)/2} \prod_{j=1}^{p} \big(q_{g(j)} + \eta\, u_j\, b_{g(j)}\big)^{1/2} \exp\left\{-\frac{1}{\xi\eta} - \sum_{j=1}^{p} \frac{\beta_j^2}{2\eta\, u_j\, b_{g(j)}}\right\}, \qquad \eta > 0. \tag{122}$$

**Allowance-induced slab variance $q_g = c_g^2$.** From (112), for each group $g$,

$$\pi(q_g \mid \text{rest}) \propto q_g^{\alpha_\kappa-1-p_g/2}(\sigma^2 + q_g)^{-(\alpha_\kappa+\beta_\kappa)} \prod_{j\in\mathcal{G}_g} \big(q_g + \eta\, u_j\, b_g\big)^{1/2} \exp\left(-\frac{\|\boldsymbol{\beta}_g\|_2^2}{2q_g}\right), \qquad q_g > 0. \tag{123}$$

**Noise variance $\sigma^2$.** Under the $q_g$-parameterization, $\sigma^2$ enters through the likelihood, its inverse-gamma prior, the induced prior on each $q_g$, and the second-stage global auxiliary prior for $\xi$. Let

$$\text{RSS}(\boldsymbol{\beta}) := \|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2.$$

Then

$$\pi(\sigma^2 \mid \text{rest}) \propto (\sigma^2)^{G\beta_\kappa - a_\sigma - \frac{n}{2} - \frac{3}{2}} \exp\left\{-\frac{1}{\sigma^2}\left(b_\sigma + \frac{\text{RSS}(\boldsymbol{\beta})}{2} + \frac{1}{\tilde{\tau}_0^2\xi}\right)\right\} \prod_{g=1}^{G} (\sigma^2+q_g)^{-(\alpha_\kappa+\beta_\kappa)}, \qquad \sigma^2 > 0. \tag{124}$$

**Proposition A.2** (Propriety of the one-dimensional conditional kernels)**.** *Fix admissible values of the remaining variables in the computational specialization of Appendix A. Then the following statements hold.*

***(i) Automatically proper kernels.*** *The conditional kernels in* (120)*,* (122)*, and* (124) *are integrable on* $(0, \infty)$*. Hence they define proper one-dimensional conditional distributions.*

***(ii) Group allowance-induced slab variance*** $q_g$***.*** *For the kernel* (123)*:*

- *if* $\|\boldsymbol{\beta}_g\|_2 > 0$*, then the kernel is proper on* $(0, \infty)$*;*
- *if* $\boldsymbol{\beta}_g = \mathbf{0}$*, then the kernel is proper on* $(0, \infty)$ *if and only if*

$$\alpha_\kappa > \frac{p_g}{2}. \tag{125}$$

***(iii) Group amplitude variance*** $b_g$***.*** *For the kernel* (121)*:*

- *if* $\|\boldsymbol{\beta}_g\|_2 > 0$*, then the kernel is proper on* $(0, \infty)$*;*
- *if* $\boldsymbol{\beta}_g = \mathbf{0}$*, then the kernel is improper near* $b_g = 0$*.*

*Proof.* We analyze the behavior of each kernel near 0 and $\infty$.

**Step 1: the kernel for $u_j$.** From (120),

$$\pi(u_j \mid \text{rest}) \propto u_j^{-2}(q_g + \eta b_g u_j)^{1/2} \exp\left\{-\frac{1}{u_j}\left(\frac{\beta_j^2}{2\eta b_g} + \frac{1}{\nu_j}\right)\right\}.$$

As $u_j \downarrow 0$, the factor inside the exponential is strictly positive because $\nu_j > 0$, so the kernel is dominated by

$$u_j^{-2} \exp(-c/u_j)$$

for some $c > 0$, which is integrable near 0. As $u_j \to \infty$,

$$(q_g + \eta b_g u_j)^{1/2} \asymp u_j^{1/2},$$

hence the kernel behaves like $u_j^{-3/2}$, which is integrable at infinity. Thus (120) is proper.

**Step 2: the kernel for $\eta$.** From (122),

$$\pi(\eta \mid \text{rest}) \propto \eta^{-(p+3)/2} \prod_{j=1}^{p} (q_{g(j)} + \eta u_j b_{g(j)})^{1/2} \exp\left\{-\frac{1}{\xi\eta} - \sum_{j=1}^{p} \frac{\beta_j^2}{2\eta u_j b_{g(j)}}\right\}.$$

As $\eta \downarrow 0$, the exponential term is bounded above by $\exp(-c/\eta)$ for some $c > 0$, hence the kernel is integrable near 0. As $\eta \to \infty$,

$$\prod_{j=1}^{p} (q_{g(j)} + \eta u_j b_{g(j)})^{1/2} \asymp \eta^{p/2},$$

so the kernel behaves like $\eta^{-3/2}$, which is integrable at infinity. Thus (122) is proper.

**Step 3: the kernel for $\sigma^2$.** From (124),

$$\pi(\sigma^2 \mid \text{rest}) \propto (\sigma^2)^{G\beta_\kappa - a_\sigma - \frac{n}{2} - \frac{3}{2}} \exp\left\{-\frac{1}{\sigma^2}\left(b_\sigma + \frac{\text{RSS}(\boldsymbol{\beta})}{2} + \frac{1}{\tilde{\tau}_0^2 \xi}\right)\right\} \prod_{g=1}^{G} (\sigma^2 + q_g)^{-(\alpha_\kappa + \beta_\kappa)}.$$

As $\sigma^2 \downarrow 0$, the coefficient inside the exponential is strictly positive because $b_\sigma > 0$ and $\xi > 0$, so the kernel is integrable near 0. As $\sigma^2 \to \infty$,

$$\prod_{g=1}^{G} (\sigma^2 + q_g)^{-(\alpha_\kappa + \beta_\kappa)} \asymp (\sigma^2)^{-G(\alpha_\kappa + \beta_\kappa)},$$

hence the kernel behaves like

$$(\sigma^2)^{-G\alpha_\kappa - a_\sigma - \frac{n}{2} - \frac{3}{2}},$$

whose exponent is strictly less than $-1$. Therefore (124) is proper.

**Step 4: the kernel for $q_g$.** From (123),

$$\pi(q_g \mid \text{rest}) \propto q_g^{\alpha_\kappa - 1 - p_g/2} (\sigma^2 + q_g)^{-(\alpha_\kappa + \beta_\kappa)} \prod_{j \in \mathcal{G}_g} (q_g + \eta u_j b_g)^{1/2} \exp\left( -\frac{\|\boldsymbol{\beta}_g\|_2^2}{2q_g} \right).$$

As $q_g \to \infty$,

$$(\sigma^2 + q_g)^{-(\alpha_\kappa + \beta_\kappa)} \asymp q_g^{-(\alpha_\kappa + \beta_\kappa)}, \qquad \prod_{j \in \mathcal{G}_g} (q_g + \eta u_j b_g)^{1/2} \asymp q_g^{p_g/2},$$

so the kernel behaves like

$$q_g^{-\beta_\kappa - 1},$$

which is integrable at infinity because $\beta_\kappa > 0$.

As $q_g \downarrow 0$, the factors

$$(\sigma^2 + q_g)^{-(\alpha_\kappa + \beta_\kappa)} \quad \text{and} \quad \prod_{j \in \mathcal{G}_g} (q_g + \eta u_j b_g)^{1/2}$$

remain bounded away from 0 and $\infty$. Hence

$$\pi(q_g \mid \text{rest}) \asymp q_g^{\alpha_\kappa - 1 - p_g/2} \exp\left( -\frac{\|\boldsymbol{\beta}_g\|_2^2}{2q_g} \right), \qquad q_g \downarrow 0.$$

If $\|\boldsymbol{\beta}_g\|_2 > 0$, the exponential term makes the kernel integrable near 0. If $\boldsymbol{\beta}_g = \mathbf{0}$, the kernel is integrable near 0 if and only if

$$\alpha_\kappa - 1 - \frac{p_g}{2} > -1,$$

that is, if and only if $\alpha_\kappa > p_g/2$. This proves (125).

**Step 5: the kernel for $b_g$.** From (121),

$$\pi(b_g \mid \text{rest}) \propto b_g^{-(p_g+1)/2} \prod_{j \in \mathcal{G}_g} (q_g + \eta u_j b_g)^{1/2} \exp\left\{ -\frac{b_g}{2s_{a,g}^2} - \sum_{j \in \mathcal{G}_g} \frac{\beta_j^2}{2\eta u_j b_g} \right\}.$$

As $b_g \to \infty$,

$$\prod_{j \in \mathcal{G}_g} (q_g + \eta u_j b_g)^{1/2} \asymp b_g^{p_g/2},$$

so the polynomial part behaves like $b_g^{-1/2}$, and the factor $\exp(-b_g/(2s_{a,g}^2))$ guarantees integrability at infinity.

As $b_g \downarrow 0$,

$$\pi(b_g \mid \text{rest}) \asymp b_g^{-(p_g+1)/2} \exp\left( -\sum_{j \in \mathcal{G}_g} \frac{\beta_j^2}{2\eta u_j b_g} \right).$$

If $\|\boldsymbol{\beta}_g\|_2 > 0$, then the exponential factor has the form $\exp(-c/b_g)$ with $c > 0$, so the kernel is integrable near 0. If $\boldsymbol{\beta}_g = \mathbf{0}$, the kernel behaves like $b_g^{-(p_g+1)/2}$, which is not integrable at 0 for any $p_g \geq 1$. This proves the claim. □

**Remark A.3.** *Proposition A.2 should be read as a statement about slice-wise conditional propriety rather than joint posterior propriety. In particular, the $q_g$- and $b_g$-parameterizations are not globally Gibbs-closed across the entire state space. This does not invalidate the primitive GR-RHS model, nor does it by itself imply impropriety of the joint posterior. Its main implication is computational: transformed HMC/NUTS or non-centered implementations are preferable to a purely Gibbs-based strategy for the full hierarchy.*

**Remark A.4.** *Under the flat allowance prior $\kappa_g \sim \mathrm{Beta}(1,1)$, one has $\alpha_\kappa = 1$. Hence the $q_g$-kernel at the zero block fails to be conditionally proper whenever $p_g \geq 2$. The same threshold appears if one works directly in the primitive $\kappa_g$-parameterization, since for $\boldsymbol{\beta}_g = \mathbf{0}$ the primitive conditional kernel behaves like*

$$\kappa_g^{\alpha_\kappa - 1 - p_g/2}$$

*near $\kappa_g = 0$. Thus this slice-wise conditional impropriety is not a pathology unique to the $q_g$ reparameterization; rather, it reflects the geometry of conditioning on an exact zero coefficient block under the multiplicative hierarchy. By itself it does not imply impropriety of the primitive model or of the full joint posterior.*

### A.4.3 Direct primitive-kernel remark for $\kappa_g$

Although $q_g$ is the more convenient computational variable, one may also write the exact full conditional kernel directly in the primitive $\kappa_g$-parameterization:

$$\pi(\kappa_g \mid \mathrm{rest}) \propto \kappa_g^{\alpha_\kappa - 1}(1-\kappa_g)^{\beta_\kappa - 1} \prod_{j \in \mathcal{G}_g} v_{j,g}^{-1/2} \exp\left(-\frac{\beta_j^2}{2v_{j,g}}\right), \qquad 0 < \kappa_g < 1, \tag{126}$$

where

$$v_{j,g} = \frac{\sigma^2 \kappa_g \, \eta \, u_j \, b_g}{\sigma^2 \kappa_g + (1-\kappa_g)\eta \, u_j \, b_g}.$$

Equations (123) and (126) are completely equivalent under the one-to-one transformation $q_g = \sigma^2 \kappa_g / (1-\kappa_g)$.

**Remark A.5.** *When $\boldsymbol{\beta}_g = \mathbf{0}$, the primitive kernel* (126) *behaves near $\kappa_g = 0$ like*

$$\kappa_g^{\alpha_\kappa - 1 - p_g/2},$$

*because $v_{j,g} \sim \sigma^2 \kappa_g$ as $\kappa_g \downarrow 0$. Hence the primitive conditional kernel is locally integrable at 0 if and only if $\alpha_\kappa > p_g/2$, exactly matching the threshold in Proposition A.2 for the transformed $q_g$-kernel.*

## A.5 Non-centered representation

A numerically convenient reparameterization is obtained by introducing standard-normal latent coefficients $z_j$ through

$$\beta_j = \sqrt{v_j}\, z_j, \qquad z_j \sim \mathcal{N}(0,1),$$

where $v_j$ is given by (110). In vector form,

$$\boldsymbol{\beta} = D_v^{1/2}\boldsymbol{z}, \qquad \boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, I_p).$$

This non-centered form does not modify the model. It is an exact reparameterization of the same hierarchy and is often preferable in HMC/NUTS implementations because it weakens some of the funnel-type posterior geometry in weakly identified regions.

## A.6 Implementation remarks

The results above show that the GR-RHS hierarchy preserves the key computational feature of regularized horseshoe-type priors: conditional on the latent scales, the coefficient block remains Gaussian. This makes block updates for $\boldsymbol{\beta}$ straightforward and provides a natural foundation for either Gaussian-block Gibbs updates or gradient-based samplers.

At the same time, several one-dimensional latent updates are nonstandard under the present allowance layer. For this reason, the full conditional kernels in (120), (121), (122), (123), and (124) are best viewed as explicit descriptions of the conditional structure rather than as a prescription to use only Gibbs sampling. In practice, we recommend transformed, preferably non-centered, HMC/NUTS as the primary implementation route. The one-dimensional kernels derived above remain useful for exposing the computational anatomy of the hierarchy and for designing auxiliary samplers where needed.

The key point is structural: the auxiliary-variable representations and transformed quantities introduced in this appendix are exact computational devices. They do not alter the primitive GR-RHS model stated in Section 1. At the same time, the appendix does not provide a general theorem of full joint posterior propriety for the complete hierarchy under arbitrary grouped designs. The results here should therefore be read as exact conditional-structure statements rather than as a proof that the full model is globally Gibbs-closed.